TODO:

DO VAL BY FILE! FIND OUTLIERS!!!!

SETUP CACHE BY
vi /usr/local/sbin/setup-dm-cache.sh

Frequencies
O4 is 5.766 / 5.838 / 5.804
Runcam is 5.839
Bee is 5.77

Log

ARTIFACTS
Intermediate freq seems to help if it gets larger, maybe it should be set proportional to something?

https://lucid.app/lucidchart/7a0fb98e-6eae-4f20-8818-dbb6c5a26e8a/edit?viewport_loc=-1710%2C-1263%2C3579%2C1779%2C0_0&invitationId=inv_beca382b-2829-4d39-b468-f1f9ca44def8
ON REBOOT
cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor
sudo /usr/bin/cpupower frequency-set -g performance

TODO
WIFI TESTING
Log longer buffer before increasing sampling rate
Does this help?
ESP32 wifi testing (at home before on rover)
Get NN training working with V2 data
Calculate size of SD card for 24 hour of recordings
PI Imaging steps
PI needs GPS time to synchronize , time sync?
Obstacle avoidance / stop
Non amp version not working
/ stop

Multipoint telemetry
https://risc.readthedocs.io/2-multi-point-telemetry.html

https://docs.holybro.com/radio/sik-telemetry-radio-v3/multiple-point-to-point-setup-with-sik-radio

https://github.com/ArduPilot/SiK/issues/44

Add avoidance sensor holder for front , and hole for wire to pi
Maia SDR FPGA example

https://www.youtube.com/watch?v=P2kChvtPxyw
Crab drone bv

Printing
JLC3DP

TEST DRONE CONFIG
python spf/spf/mavlink_radio_collection.py --yaml-config ./spf/spf/rover_configs/rover_receiver_config_pi.yaml -m /home/pi/device_mapping -r bounce -t "RO1" -n 40 --drone-uri tcp:192.168.1.141:14590 --no-ultrasonic

Zarr_fix
for x in *.zarr; do if [! -d "../june_fix/$x"]; then python ~/gits/spf/spf/dataset/zarr_rechunk.py $x ../june_fix/$x; fi; done

RSYNC
while [1 -lt 2]; do rsync -avP --remove-source-files pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.zarr' ./ ; rsync -avP pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.yaml' ./ ; sleep 600; done

rsync -av --include="*/" --include="best.pth" --include="empirical_dist.pkl" --include="config.yml" --exclude="*" /home/mouse9911/gits/spf/checkpoints /mnt/md2/cache/inference_models/

B2 upload

ANALOG DOCS

https://wiki.analog.com/resources/eval/user-guides/ad-fmcomms1-ebz/math
https://wiki.analog.com/university/tools/pluto/users/non_quad

Other docs

https://gigayasawireless.github.io/5G-on-SDRs/index.html

TODO
#Save mean_phase in precomputed , avoid pkl in temp files to avoid O(n^2) writes
Only compute beamformer for the segmented windows

Dotenv .env
B2_ENDPOINT = 'https://s3.us-west-004.backblazeb2.com'
B2_KEY_ID='004bd519e54c8390000000003'
B2_APP_KEY='K004cvp6xJpJs1W9ogA7sidXim/WPwo'

B2 SYNC
while [1 -lt 2]; do date ; b2 sync --delete /mnt/md2/ b2://projectspf/md2/; b2 sync --delete /mnt/md1/ b2://projectspf/md1/; b2 sync --delete /mnt/md0/ b2://projectspf/md0/; echo done; sleep 9000; done

December 15
python spf/scripts/train_single_point.py -c model_configs/jul8_single_3p7_thin_noblade_dec15.yaml

Jul 13
grep -i blade /mnt/md2/cache/nosig_data/*.yaml | awk '{print $1}' | sed 's/://g' | uniq | sed 's/.yaml/.zarr/g' > bladerf_files.txt

July 8
Missing ls -d /mnt/md[012]/2d_wallarray_v2_data/*/*.zarr /mnt/md2/rovers/merged/*.zarr | while read x ; do z=`echo $x | sed 's/.*wallarray/wallarray/g' | sed 's@.*/merged/@@' | sed 's/.zarr/_segmentation_nthetas65.yarr/g'` ; zz="/mnt/md2/cache/precompute_cache_3p7/$z/data.mdb"; if [! -f "$zz"] ; then echo $zz $x; fi ; done

export DISPLAY=:0
python spf/spf/mavlink_radio_collection_inference.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic --ignore-mode --no-write-to-disk --checkpoint ~/checkpoints/paired_3p7_thinner/best.pth --checkpoint-config ~/checkpoints/paired_3p7_thinner/config.yml --realtime

June 11

python spf/spf/mavlink_radio_collection_inference.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic --ignore-mode --no-write-to-disk --checkpoint ~/checkpoints/paired_3p7_thinner/best.pth --checkpoint-config ~/checkpoints/paired_3p7_thinner/config.yml --realtime

June 10
python spf/spf/mavlink_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic --ignore-mode --no-write-to-disk --checkpoint ~/checkpoints/paired_3p7_thinner/best.pth --checkpoint-config ~/checkpoints/paired_3p7_thinner/config.yml --realtime

June 9
python spf/spf/mavlink_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic --ignore-mode --no-write-to-disk --checkpoint ~/checkpoints/paired_3p6_thinner/best.pth --checkpoint-config ~/checkpoints/paired_3p6_thinner/config.yml --realtime
NEEDS TO BE PAIRED

python spf/spf/mavlink_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic --ignore-mode --no-write-to-disk --checkpoint ~/checkpoints/single_3p7_thinner_rerun/best.pth --checkpoint-config ~/checkpoints/single_3p7_thinner_rerun/config.yml --realtime

June 4
f

April27
py-spy top -- python spf/spf/mavlink_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic --ignore-mode --no-write-to-disk --checkpoint ~/checkpoints/thinner/best.pth --checkpoint-config ~/checkpoints/thinner/config.yml --realtime

April 26:
python spf/spf/mavlink_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic --ignore-mode

python spf/spf/mavlink_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping

--no-ultrasonic --ignore-mode --no-write-to-disk --checkpoint ~/checkpoints/thinner/best.pth
--checkpoint-config ~/checkpoints/thinner/config.yml

python spf/spf/mavlink_radio_collection.py --yaml-config
spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping
--no-ultrasonic --ignore-mode --no-write-to-disk --checkpoint ~/checkpoints/thinner/best.pth
--checkpoint-config ~/checkpoints/thinner/config.yml --realtime

Compile pytorch rpi5

```
sudo apt install -y \
  git cmake ninja-build \
  python3-dev python3-pip \
  libopenblas-dev libblas-dev \
  liblapack-dev libprotobuf-dev protobuf-compiler \
  build-essential libjpeg-dev zlib1g-dev libomp-dev

cd ~
git clone --recursive https://github.com/pytorch/pytorch
cd pytorch
git checkout v2.3.0
git submodule sync
git submodule update --init --recursive

# Compiler flags to use NEON and VFPv4 on Pi5 (Cortex-A76/A72 cores)
export USE_NCCL=0
export USE_CUDA=0
export USE_XNNPACK=1
export USE_OPENMP=1
export USE_NNPACK=1
export BUILD_CAFFE2=0
export TORCH_CUDA_ARCH_LIST=""

# Special ARM optimizations:
export CFLAGS="-mcpu=cortex-a72+nocrypto"
export CXXFLAGS="-mcpu=cortex-a72+nocrypto"

export MAX_JOBS=$(nproc)  # Use all cores

pip install wheel
python3 setup.py bdist_wheel
```

April 25

python spf/spf/mavlink_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic --ignore-mode

April 24

python spf/spf/grbl_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/rpi5_capture_5866M_35mm.yaml -r bounce -o /tmp/test6 --device-mapping ~/device_mapping --n-records 600

python spf/spf/mavlink_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/inference_config.yaml --device-mapping ~/device_mapping --no-ultrasonic

April 21

System crash, brought up with /etc/systemd/system/dm-cache-setup.service

Warm cache

while [1 -lt 2]; do cat /mnt/md2/cache/nosig_data/*/* /mnt/md2/cache/precompute_cache_3p6/* /mnt/md2/cache/precompute_cache_3p6/*/* > /dev/null; sleep 600 ; done

April 14

https://grabcad.com/library/5-inch-capacitive-touch-ips-display-800x480-mipi-dsi-1

Freenove 5inch dsi display

April 13

Generate new classic token

#docker build . -t ardupilotsim:v0.1

docker build . -t ghcr.io/misko/ardupilotspf:v0.2

docker push ghcr.io/misko/ardupilotspf:v0.2

docker pull ghcr.io/misko/ardupilotspf:v0.2

April 8

Iio_info on osx

sudo /Library/Frameworks/iio.framework/Tools/iio_attr /Library/Frameworks/iio.framework/Tools/iio_info -S

python spf/spf/grbl_radio_collection.py --yaml-config spf/data_collection/rpi5_inference/rpi5_capture_5866M_35mm.yaml -r bounce -o /tmp/test1 --device-mapping ~/device_mapping

April 6

rsync -avP /mnt/md2/2d_wallarray_v2_data/jan /mnt/md0/2d_wallarray_v2_data/

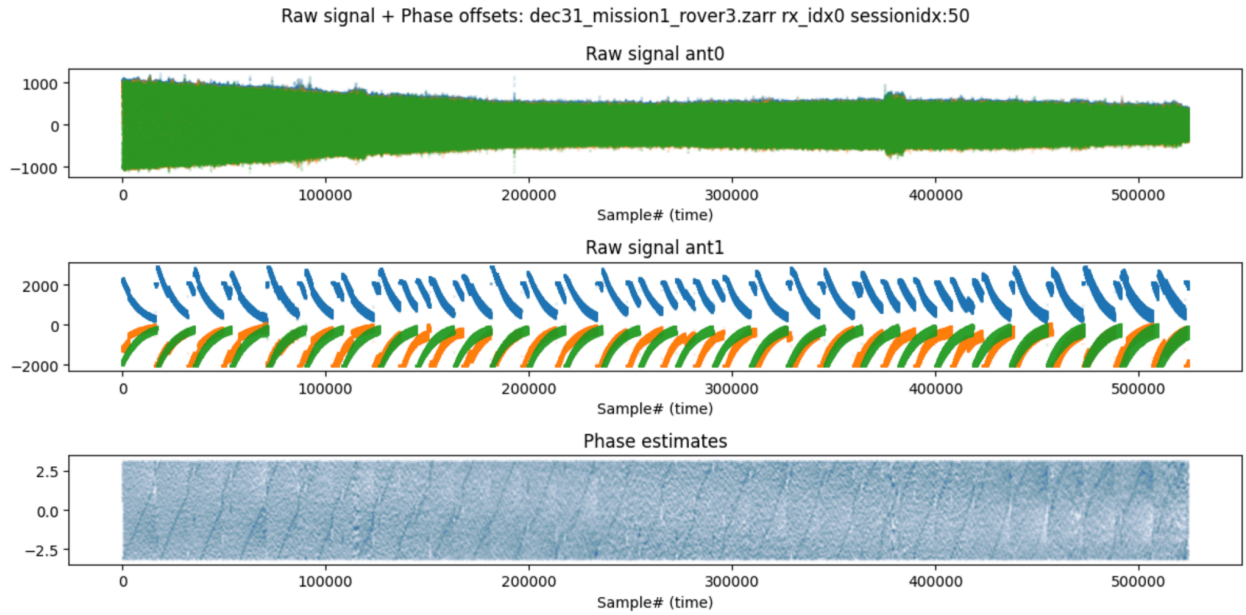
DFU on HAMGEEK Plutoplus sdr
Set jumper URST + MIO52 not URST + MIO46 (on other boards), hold DFU
Green light should come on and stay on, try MIO52<>URST or MIO46<>URST if not working
When booted sudo dfu-util -l
sudo dfu-util -a firmware.dfu -D pluto.dfu
sudo `dfu-util -a boot.dfu -D ./boot.dfu`
sudo `dfu-util -a uboot-env.dfu -D ./uboot-env.dfu`

`UNCLEAR WHY BUT newer timestamp 0.38 firmware does not work on some PLUTOPLUS`

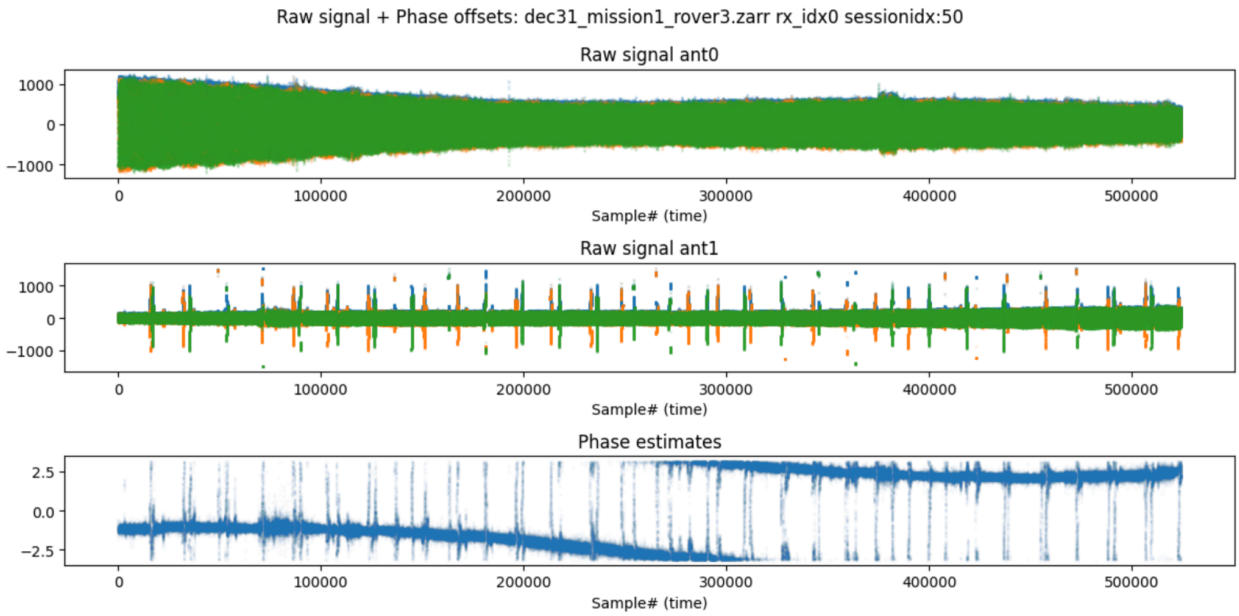
Segmentation v3.6
 ls -d /mnt/md[012]/2d_wallarray_v2_data/*/*.zarr /mnt/md2/rovers/merged/*.zarr | shuf | while read x ; do python ~/gits/spf/spf/scripts/segment_zarr.py -i $x --precompute-cache /mnt/md2/cache/precompute_cache_3p6/ -w 0 -p 16; done

Windows 11 virtual machine telemetry ft 230 basic uart adapter
Need to manually install FTDI drivers from site for ARM**
 testing chatgpt fast detrend

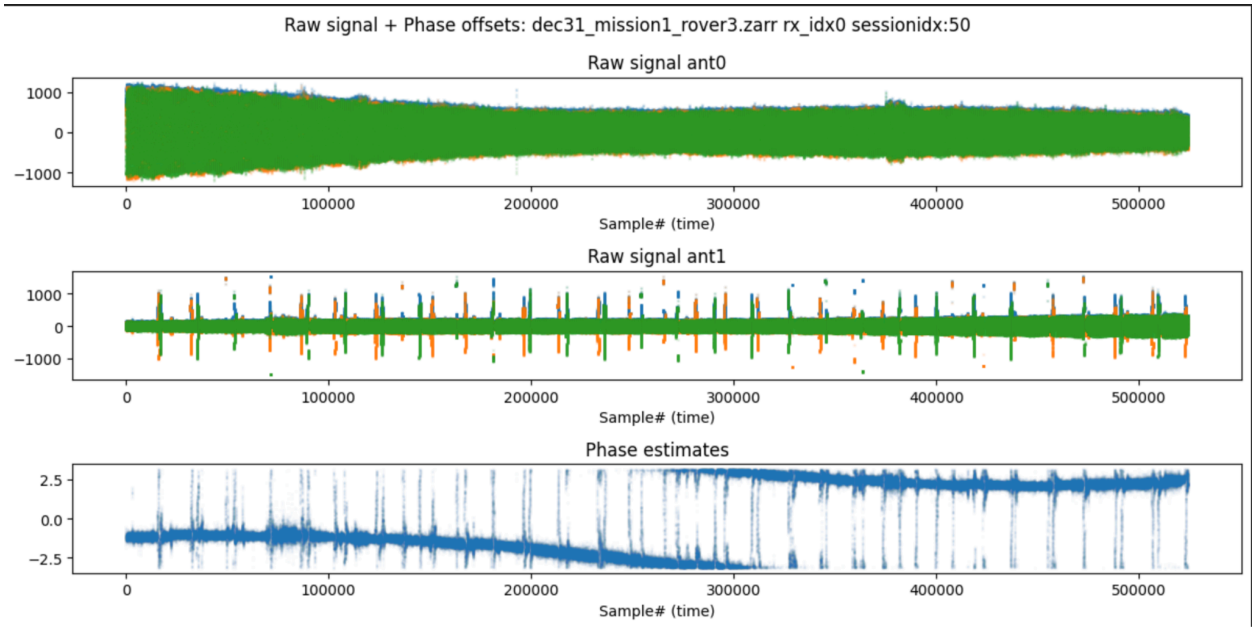
Without detrend



With old detrend



With new detrend



April 5
Evaluate half data and quarter data rover held out?

Create empirical dist
python spf/scripts/create_empirical_p_dist.py -d /mnt/md2/cache/nosig_data/*.zarr --precompute-cache /mnt/md2/cache/precompute_cache_3p5_chunk1/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

Configure lowest power setting on rover. Seem to get stuck near waypoints. Not moving

Replace all power switches faulty
Rover 2 compass calibration through mavproxy
Magcal start and accept

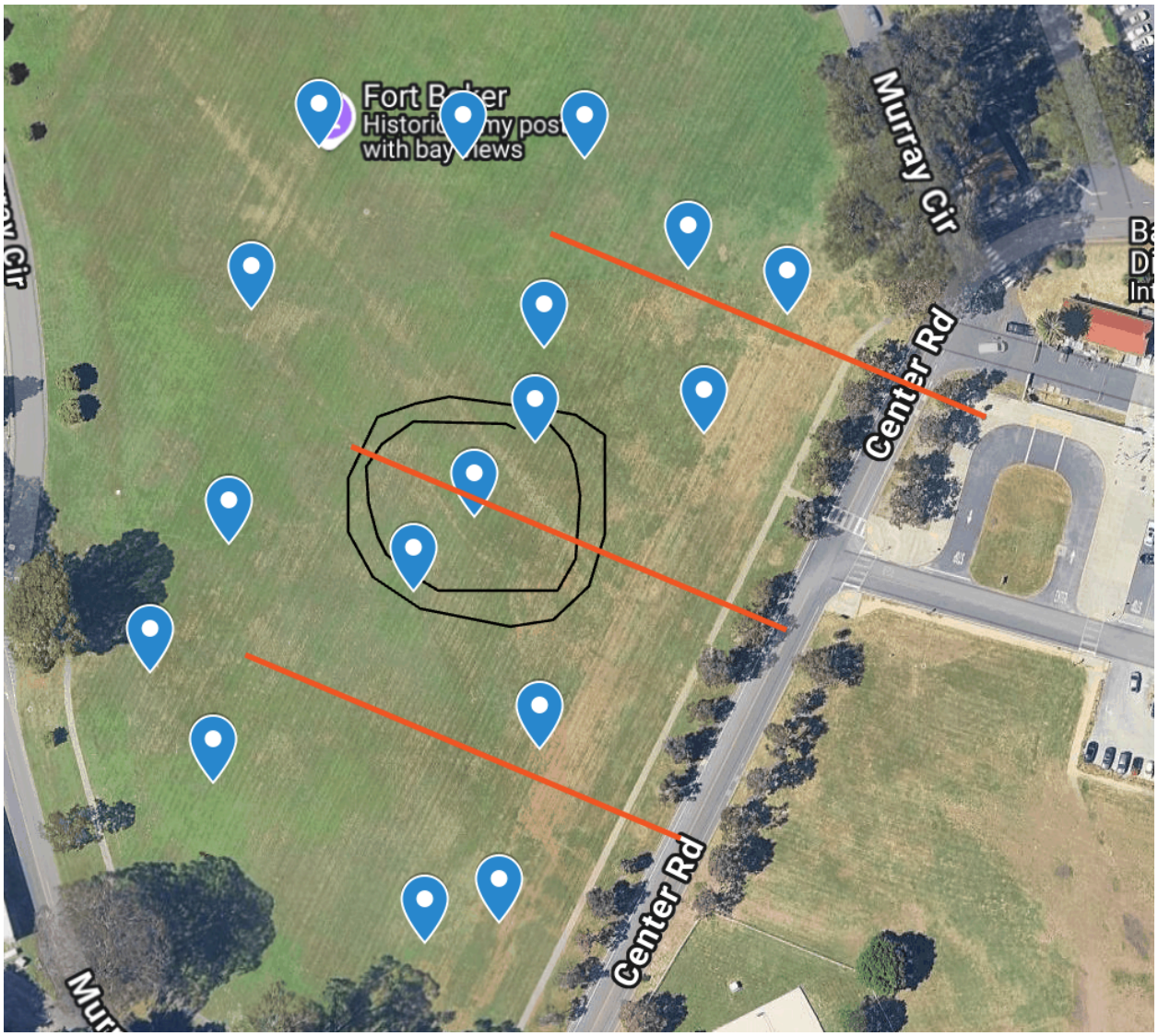
python spf/scripts/v4_tx_rx_to_v5.py --txs /mnt/md2/rovers/rover2/rover_2025_04_05_*.zarr --rxs /mnt/md2/rovers/rover1/rover_2025_04_05*.zarr --output /mnt/md2/rovers/merged_test/
python spf/scripts/v4_tx_rx_to_v5.py --txs /mnt/md2/rovers/rover2/rover_2025_04_05_*.zarr --rxs /mnt/md2/rovers/rover3/rover_2025_04_05*.zarr --output /mnt/md2/rovers/merged_test/

April 4

python spf/spf/mavlink_radio_collection.py -c spf/data_collection/rover/rover_v3.1/capture_configs/rover_single_receiver_config_pi_3mhz_35mm.yaml -m /home/pi/device_mapping -r circle -t "RO2" -n 100 --drone-uri tcp:192.168.1.141:14591 --no-ultrasonic

python spf/spf/mavlink_radio_collection.py -c spf/data_collection/rover/rover_v3.1/capture_configs/rover_receiver_config_pi_3mhz_35mm.yaml -m /home/pi/device_mapping -r circle -t "RO3" -n 100 --drone-uri tcp:192.168.1.141:14591 --no-ultrasonic

python spf/spf/mavlink_radio_collection.py -c spf/data_collection/rover/rover_v3.1/capture_configs/rover_receiver_config_pi_3mhz_35mm.yaml -m /home/pi/device_mapping -r circle -t "RO1" -n 100 --drone-uri tcp:192.168.1.141:14591 --no-ultrasonic



Fort baker right

docker run --rm -it -p 14590-14595:14590-14595 csmisko/ardupilotspf:latest /ardupilot/Tools/autotest/sim_vehicle.py -l 37.835940,-122.478244,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590 --out tcpin:0.0.0.0:14591

docker run --rm -it -p 14590-14595:14590-14595 csmisko/ardupilotspf:latest /ardupilot/Tools/autotest/sim_vehicle.py -l 37.835450,-122.478590,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590 --out tcpin:0.0.0.0:14591

docker run --rm -it -p 14590-14595:14590-14595 csmisko/ardupilotspf:latest /ardupilot/Tools/autotest/sim_vehicle.py -l 37.834975,-122.478842,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590 --out tcpin:0.0.0.0:14591

,

April 1
https://www.youtube.com/watch?v=ghhqSGxdIwQ&t=227s
OSD microOSD

March 30
Met with Jason + data collection

> See if beamformer in one rf window matters over time, or if its similar enough to the beamformer over the whole 512k window (vs 256 windows)
> Run network on averaged beam former to see if results are the same
>
> Multiply single outputs out

March 29
python spf/scripts/v4_tx_rx_to_v5.py --txs /mnt/md2/rovers/rover2/rover_2025_03_29_*.zarr --rxs /mnt/md2/rovers/rover1/rover_2025_03_29*.zarr --output /mnt/md2/rovers/merged_test/
python spf/scripts/v4_tx_rx_to_v5.py --txs /mnt/md2/rovers/rover2/rover_2025_03_29_*.zarr --rxs /mnt/md2/rovers/rover3/rover_2025_03_29*.zarr --output /mnt/md2/rovers/merged_test/

March 24
Try half data in training only to see affect on val (unchanged)

March 22
TESTING
docker run --rm -it -p 14590-14595:14590-14595 csmisko/ardupilotspf:latest /ardupilot/Tools/autotest/sim_vehicle.py -l 37.76509485,-122.40910127,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590 --out tcpin:0.0.0.0:14591

python spf/spf/mavlink_radio_collection.py -c ${config} -m /home/pi/device_mapping -r circle -t "RO2" -n 100 --drone-uri tcp:192.168.1.141:14591 --no-ultrasonic

python spf/scripts/v4_tx_rx_to_v5.py --txs /mnt/md2/rovers/rover2/rover_2025_02_23_*.zarr --rxs /mnt/md2/rovers/rover1/rover_2025_02_23_*.zarr --output /mnt/md2/rovers/merged_test/

python spf/scripts/v4_tx_rx_to_v5.py --txs /mnt/md2/rovers/rover2/rover_2025_03_22_*.zarr --rxs /mnt/md2/rovers/rover1/rover_2025_03_22_*.zarr --output /mnt/md2/rovers/merged_test/;
python spf/scripts/v4_tx_rx_to_v5.py --txs /mnt/md2/rovers/rover2/rover_2025_03_22_*.zarr --rxs /mnt/md2/rovers/rover3/rover_2025_03_22_*.zarr --output /mnt/md2/rovers/merged_test/

March 21
Able to run 192 cores on AWS , its a bit finicky , you have to wait 10-15min between instances
192 cores runs within 50GB RAM

python ~/gits/spf/spf/scripts/run_filters_on_data_b2.py --config ~/gits/spf/spf/model_training_and_inference/models/ekf_and_pf_config.yml --empirical-pkl-fn

b2://projectspf/md2/cache/inference_models/checkpoints/march16/paired_wd0p02_gains_vehicle_0p2dropout_noroverbounceREAL_lowdrop_x2/empirical_dist.pkl --parallel 28 --work-dir march16inference --manifest b2://projectspf/md2/splits/march16_val_nosig_noroverbounce.txt --seed 1337

python spf/scripts/run_filters_report_b2.py --work-dir march16inference --output-prefix /mnt/md2/reports/march16inference_x6/csv/

python ~/gits/spf/spf/scripts/run_filters_on_data_b2.py --config ~/gits/spf/spf/model_training_and_inference/models/ekf_and_pf_config.yml --empirical-pkl-fn b2://projectspf/md2/cache/inference_models/checkpoints/march16/paired_wd0p02_gains_vehicle_0p2dropout_noroverbounceREAL_lowdrop_x2/empirical_dist.pkl --parallel 28 --work-dir march16inference --manifest b2://projectspf/md2/splits/march16_val_nosig_noroverbounce.txt --debug

python ~/gits/spf/spf/scripts/run_filters_on_data_b2.py --config ~/gits/spf/spf/model_training_and_inference/models/ekf_and_pf_config.yml --empirical-pkl-fn b2://projectspf/md2/cache/inference_models/checkpoints/march16/paired_wd0p02_gains_vehicle_0p2dropout_noroverbounceREAL_lowdrop_x2/empirical_dist.pkl --parallel 28 --work-dir march16inference --manifest b2://projectspf/md2/splits/march16_val_nosig_noroverbounce.txt

March 20
docker build -t spf/spf . --no-cache

python /spf/spf/scripts/run_filters_on_data_b2.py --config ..//model_training_and_inference/models/ekf_and_pf_config.yml --empirical-pkl-fn b2://projectspf/md2/cache/inference_models/checkpoints/march16/paired_wd0p02_gains_vehicle_0p2dropout_noroverbounceREAL_lowdrop_x2/empirical_dist.pkl --parallel 28 --work-dir march16inference --manifest b2://projectspf/md2/splits/march16_val_nosig_noroverbounce.txt --debug

python /spf/spf/scripts/run_filters_on_data_b2.py --config ..//model_training_and_inference/models/ekf_and_pf_config.yml --empirical-pkl-fn b2://projectspf/md2/cache/inference_models/paired_checkpoints_windowedbeamformer_nosig_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/empirical_dist.pkl --parallel 28 --work-dir march8inference --manifest b2://projectspf/md2/splits/feb16_val_nop[

sig_with_rover.txt --debug

ECS ECR container internet subnet

Cluster needs to have public subnets and need to set get public IP
Task needs to run in bridge mode otherwise we need a NAT(?) , bridge mode works and container gets internet access

Create inference cache
d=/mnt/md2/cache/inference_models/checkpoints/march16/paired_wd0p02_gains_vehicle_0p2dropout_noroverbounceREAL_lowdrop_x2/; python spf/scripts/create_inference_cache.py --inference-cache /mnt/md2/cache/inference/ --config-fn ${d}/config.yml --checkpoint-fn ${d}/best.pth --device cuda --datasets /mnt/md2/cache/nosig_data/*.zarr --parallel 24 --precompute-cache /mnt/md2/cache/precompute_cache_3p5_chunk1/ --segmentation-version 3.5

python spf/scripts/create_empirical_p_dist.py -d /mnt/md2/cache/nosig_data/*.zarr --precompute-cache /mnt/md2/cache/precompute_cache_3p5_chunk1/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

March 18
docker build -t spf_inference:march8 . --no-cache
LOWER INPUT DROPOUT CONVERGES MUCH FASTER

docker run -it -e AWS_ACCESS_KEY_ID=AKIA2NK3X56T6POSCOWK -e AWS_SECRET_ACCESS_KEY=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 -e AWS_DEFAULT_REGION=us-west-1 spf/spf

March 17
docker run -it -e AWS_ACCESS_KEY_ID=AKIA2NK3X56T6POSCOWK -e AWS_SECRET_ACCESS_KEY=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 -e AWS_DEFAULT_REGION=us-west-1 spf:main /bin/bash
python /spf/spf/scripts/run_filters_on_data_b2.py --config ..//model_training_and_inference/models/ekf_and_pf_config.yml --empirical-pkl-fn b2://projectspf/md2/cache/inference_models/paired_checkpoints_windowedbeamformer_nosig_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/empirical_dist.pkl --parallel 28 --work-dir march8inference --manifest b2://projectspf/md2/splits/feb16_val_nosig_with_rover.txt --debug

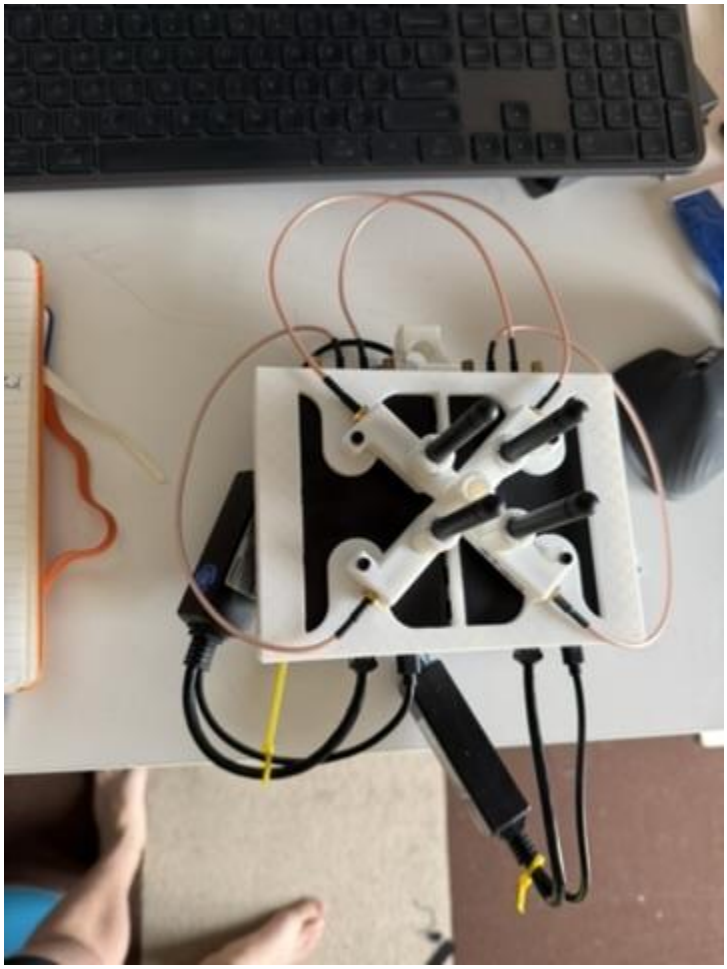
March 16

Nuand setup

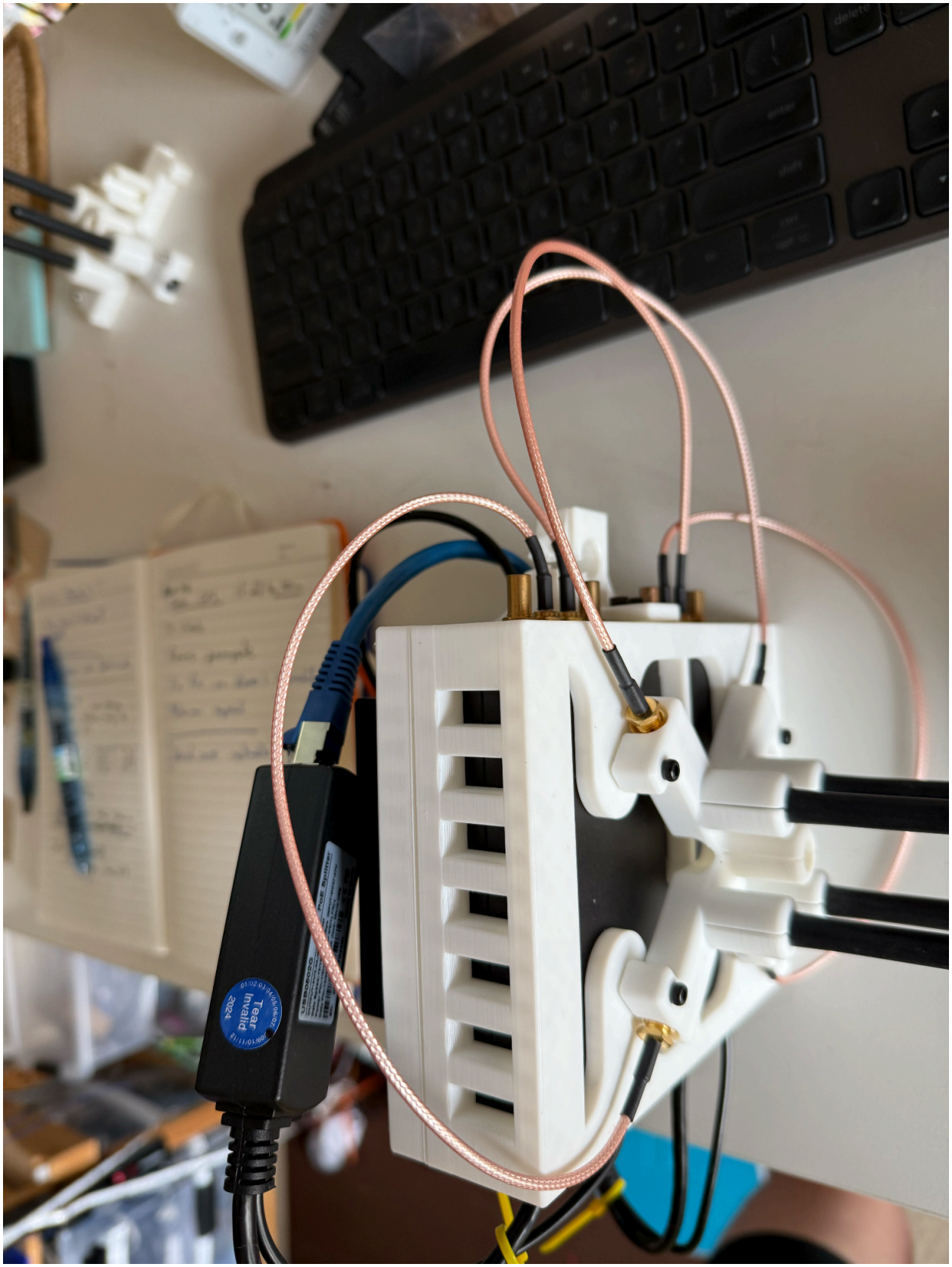
On the nuand setup the left one is 7d50f504c96a4dfdb885052d802c05db (equivalent to 192.168.1.18)

The other one is 86303349cded495984b77180464db2e8 (equivalent to 192.168.1.17)

python spf/grbl_radio_collection.py -c spf/v5_configs/wall_array_nuand_v2_external_25_o4_5766_fastattack.yaml -r bounce -s none --n-records 1000
Test setup



This is the side facing the wall
The left unit in this picture ^^ is 192.168.1.18 , the right one is 192.168.1.17


Tear
Invalid
2024

Debug gps / position data
(spf) mouse9911@kalman:/mnt/md2/rovers/rover2$ python ~/gits/spf/spf/scripts/gps_check.py -i /mnt/md2/rovers/rover2/rover_2025_02_23_23_13_54_nRX1_circle_spacing0p05075_tag_RO2.zarr
/mnt/md2/rovers/rover2/rover_2025_02_23_23_13_54_nRX1_circle_spacing0p05075_tag_RO2.zarr,r0,336,0
/mnt/md2/rovers/rover2/rover_2025_02_23_23_13_54_nRX1_circle_spacing0p05075_tag_RO2.zarr: failed 'r1'

Implement gps and position checkers

Data surgery , update rover 1 february 23 to 47mm spacing array
(spf) mouse9911@kalman:/mnt/md2/rovers/rover1$ grep 0.043 rover_2025_02_23*.yaml
rover_2025_02_23_17_45_15_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_17_45_15_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_18_25_21_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_18_25_21_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_18_56_41_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_18_56_41_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_21_31_35_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_21_31_35_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_22_04_54_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_22_04_54_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_22_40_06_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_22_40_06_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_23_14_18_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_23_14_18_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043
rover_2025_02_23_23_46_11_nRX2_diamond_spacing0p043_tag_RO1.yaml:-antenna-spacing-m: 0.043

rover_2025_02_23_23_46_11_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.043
(spf) mouse9911@kalman:/mnt/md2/rovers/rover1$ grep 0.043 rover_2025_02_23*.yaml^C
(spf) mouse9911@kalman:/mnt/md2/rovers/rover1$ sed -i 's/0.043/0.047/g'
rover_2025_02_23*.yaml
(spf) mouse9911@kalman:/mnt/md2/rovers/rover1$ grep spacing rover_2025_02_23*.yaml
rover_2025_02_23_17_45_15_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_17_45_15_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_18_25_21_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_18_25_21_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_18_56_41_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_18_56_41_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_21_31_35_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_21_31_35_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_22_04_54_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_22_04_54_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_22_40_06_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_22_40_06_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_23_14_18_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_23_14_18_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_23_46_11_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047
rover_2025_02_23_23_46_11_nRX2_diamond_spacing0p043_tag_RO1.yaml:-
antenna-spacing-m: 0.047

(spf) mouse9911@kalman:/mnt/md2/rovers/rover1$ for x in rover_2025_02_23*.zarr; do python
~/gits/spf/spf/scripts/zarr_fix_rx_spacing.py $x; done
rover_2025_02_23_17_45_15_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.0 0.043
rover_2025_02_23_18_25_21_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.043 0.043
rover_2025_02_23_18_56_41_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.043 0.043

rover_2025_02_23_21_31_35_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.043 0.043
rover_2025_02_23_22_04_54_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.043 0.043
rover_2025_02_23_22_40_06_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.043 0.043
rover_2025_02_23_23_14_18_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.043 0.043
rover_2025_02_23_23_46_11_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.043 0.043

python ~/gits/spf/spf/scripts/zarr_fix_rx_spacing.py
rover_2025_03_02_22_20_46_nRX2_diamond_spacing0p043_tag_RO1.zarr
rover_2025_03_02_22_20_46_nRX2_diamond_spacing0p043_tag_RO1.zarr 0.047 0.043 0.043

Serial over network
ACM0 https://chatgpt.com/share/67d72c2a-374c-8002-b124-3d5b0e237ece
https://wifizoo.org/2023/05/12/yet-another-ser2net-tutorial/
https://chatgpt.com/share/67d72c2a-374c-8002-b124-3d5b0e237ece

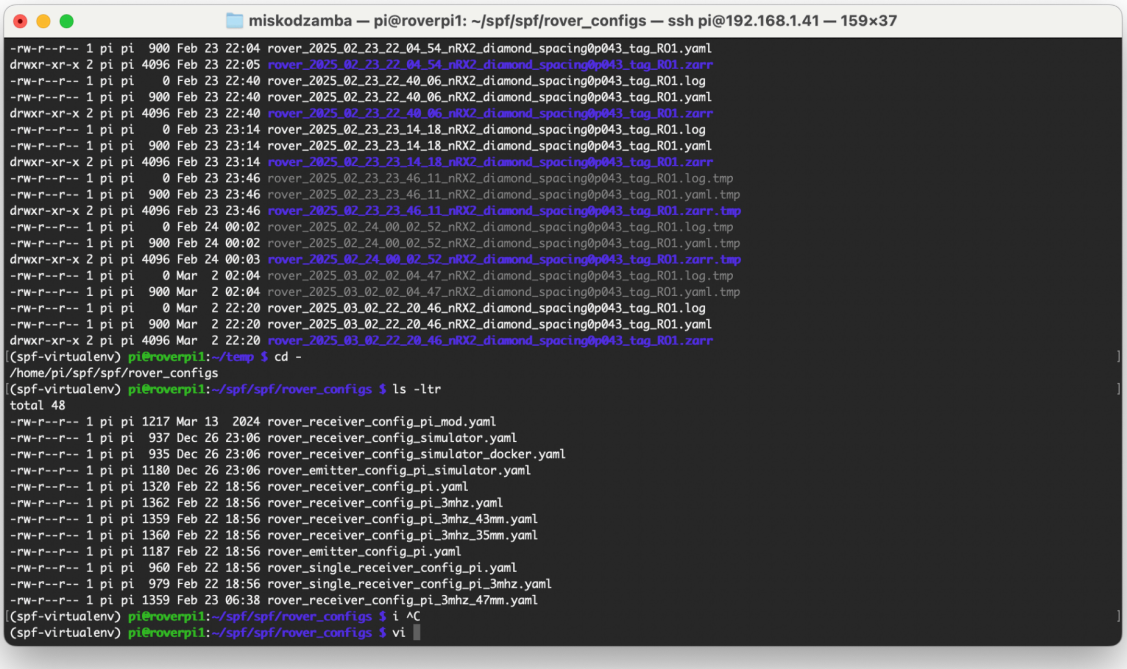
APCUPSD onbattery
pi@devpi:/etc/apcupsd $ cat onbattery

```
#!/bin/bash
#
# This shell script if placed in /etc/apcupsd
# will be called by /etc/apcupsd/apccontrol when the UPS
# goes on batteries.
# We send an email message to root to notify him.
#
#exec > /tmp/apcupsd_debug.log 2>&1
#set -x
HOSTNAME=`hostname`
MSG="$HOSTNAME UPS $1 Power Failure !!!"
#
(
  echo "$MSG"
  echo " "
  /sbin/apcaccess status
) | $APCUPSD_MAIL -s "$MSG" $SYSADMIN

systemctl stop ser2net
sleep 2

export ROOT=/home/pi/spf
source /home/pi/spf-virtualenv/bin/activate
#first need to kill off whatever processes we have
```

```
echo "KILL RUNNING CAPTURE"
ps aux | grep -v grep | grep run_capture | awk '{print $2}' | xargs kill
ps aux | grep -v grep | grep python | awk '{print $2}' | xargs kill
sleep 5
ps aux | grep -v grep | grep run_capture | awk '{print $2}' | xargs kill
ps aux | grep -v grep | grep python | awk '{print $2}' | xargs kill
sleep 5
ps aux | grep -v grep | grep run_capture | awk '{print $2}' | xargs kill
ps aux | grep -v grep | grep python | awk '{print $2}' | xargs kill
sleep 5

sleep 10

echo "WAITING TO GO HOME"
cat ${ROOT}/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/go_home |
python ${ROOT}/spf/grbl/grbl_interactive.py /dev/ttyACM0
sleep 15

echo "SHUTDOWN"
shutdown -P
exit 0
```

March 15

Fix yaml and zarr for previously recorded files that were 43mm but were actually 47mm



March 12

grep bounce /mnt/md2/cache/nosig_data/*.json | grep rover | awk '{print $1}' | sed 's/://g' | xargs -I basename | sed 's/.ls.json//g' > /mnt/md2/splits/bounce_rover.txt

cat march11_train_nosig_noroverbounce.txt | while read line; do if [[$line == *"_tag_"*]]; then echo $line 5; else echo $line ; fi done > march11_train_nosig_noroverbounce_5xrover.txt

while [1 -lt 2]; do date ; b2 sync --threads 2 ./ b2://projectspf/md2/; sleep 300; done

sudo bladeRF-cli -v verbose -d libusb:device=4:2 -L hostedxA4-latest.rbf

March 11

python run_filters_on_data_b2.py --config ..//model_training_and_inference/models/ekf_and_pf_config.yml --empirical-pkl-fn b2://projectspf/md2/cache/inference_models/paired_checkpoints_windowedbeamformer_nosig_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/empirical_dist.pkl --parallel 28 --work-dir march8inference --manifest b2://projectspf/md2/splits/feb16_val_nosig_with_rover.txt

python spf/scripts/create_empirical_p_dist.py -d /mnt/md2/cache/nosig_data/*.zarr --precompute-cache /mnt/md2/cache/precompute_cache_3p5_chunk1/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

git clone --recursive https://github.com/Nuand/bladeRF.git
https://github.com/Nuand/bladeRF/tree/master/host
bladeRF-cli -v verbose -L ~/Downloads/hostedxA4-latest.rbf
sudo bladeRF-cli -e info -e version

Wget https://www.nuand.com/fpga/hostedxA4-latest.rbf
bladeRF-cli -v verbose -L

sudo bladeRF-cli -v verbose -d libusb:device=4:2 -L hostedxA4-latest.rbf

March 9
Inference cache
python spf/scripts/create_inference_cache.py --inference-cache /mnt/md2/cache/inference/ --config-fn latest_configs/feb16_paired_wd0p02_gains_vehicle_0p2dropout_nobounce.yml --checkpoint-fn /home/mouse9911/gits/spf/checkpoints/feb16/paired_checkpoints_windowedbeamformer_nosig_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/best.pth --device cuda --datasets /mnt/md2/cache/nosig_data/*.zarr --parallel 24 --precompute-cache /mnt/md2/cache/precompute_cache_3p5_chunk1/ --segmentation-version 3.5

python run_filters_on_data_b2.py --config ..//model_training_and_inference/models/ekf_and_pf_config.yml --empirical-pkl-fn b2://projectspf/md2/cache/inference_models/paired_checkpoints_windowedbeamformer_nosig_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/empirical_dist.pkl --parallel 28 --work-dir march8inference --manifest /mnt/md2/splits/feb16_val_nosig.txt

python spf/scripts/run_filters_report_b2.py --work-dir march8inference --output-prefix /mnt/md2/reports/march8inference/csv/

March 2
python run_filters_on_data_b2.py --config ..//model_training_and_inference/models/ekf_and_pf_config_debug.yml --empirical-pkl-fn b2://projectspf/md2/cache/inference_models/paired_checkpoints_windowedbeamformer_nosig_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/empirical_dist.pkl --parallel 16 --work-dir march1inference

March 1
Amazon dynamodb
python run_filters_on_data_b2.py --config
..//model_training_and_inference/models/ekf_and_pf_config_debug.yml --empirical-pkl-fn
b2://projectspf/md2/cache/inference_models/paired_checkpoints_windowedbeamformer_nosig_
3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/empirical_dist.pkl --parallel 16
--work-dir march1inference

Feb 27
botocore.exceptions.ClientError: An error occurred (InvalidArgument) when calling the
GetObject operation: Unsupported header 'x-amz-checksum-mode' received for this API call.
(spf_venv) ubuntu@ip-172-31-29-243:~/spf/spf/scripts$
AWS_REQUEST_CHECKSUM_CALCULATION=WHEN_REQUIRED
AWS_RESPONSE_CHECKSUM_VALIDATION=WHEN_REQUIRED python boto_b2.py
Downloading md2/cache/nosig_data/dec18_mission1_rover1.zarr/data.mdb -> data.mdb

Feb 27
/etc/systemd/system/dm-cache-setup.service dmsetup cache md cache

Feb 26
sudo apt-get update
sudo apt-get install -y git screen libiio-dev libiio-utils vim python3-dev uhubctl libusb-dev
libusb-1.0-0-dev
python -m venv --system-site-packages ./spf_venv
source ./spf_venv/bin/activate
git clone https://github.com/misko/spf.git
cd spf
pip install -e .

Feb 25
#create inference cache
python spf/scripts/create_inference_cache.py --inference-cache /mnt/md2/cache/inference/
--config-fn
/home/mouse9911/gits/spf/checkpoints/feb16/paired_checkpoints_windowedbeamformer_nosig
_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/config.yml --checkpoint-fn
/home/mouse9911/gits/spf/checkpoints/feb16/paired_checkpoints_windowedbeamformer_nosig
_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/best.pth --device cuda
--datasets /mnt/md2/cache/nosig_data/*.zarr --parallel 24 --precompute-cache
/mnt/md2/cache/precompute_cache_3p5_chunk1/ --segmentation-version 3.5

#create filter outputs
cat /mnt/md2/splits/feb16_val_nosig_noroverbounce.txt | shuf | while read x; do python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc --empirical-pkl-fn `pwd`/empirical_dists/full.pkl --work-dir /mnt/md2/inference_results/feb16_workdir --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml --parallel 24; done

Feb22
Fort baker
docker run --rm -it -p 14590-14595:14590-14595 csmisko/ardupilotspf:latest /ardupilot/Tools/autotest/sim_vehicle.py -l 37.83515839511702,-122.47807074209098,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590 --out tcpin:0.0.0.0:14591

Start moving data to md1
rsync -a aug june_fix sept /mnt/md1/2d_wallarray_v2_data/

python spf/scripts/create_inference_cache.py --inference-cache /mnt/md2/cache/inference/ --config-fn latest_configs/feb16_paired_wd0p02_gains_vehicle_0p2dropout_nobounce.yml --checkpoint-fn /home/mouse9911/gits/spf/checkpoints/feb16/paired_checkpoints_windowedbeamformer_nosig_3p5_randx_wd0p02_gains_vehicle_0p2dropout_noroverbounce/best.pth --device cuda --datasets /mnt/md2/2d_wallarray_v2_data/*/*.zarr /mnt/md2/rovers/merged/*.zarr --parallel 24 --precompute-cache /mnt/md2/cache/precompute_cache_3p5_chunk1/ --segmentation-version 3.5

Feb21
DFU util unbrick
DFU Upgrade
If you device is not responding anymore, you have to apply DFU procedure to unbrick it.
1. Download the firmware and unzip it. You can get them [here](https://github.com/analogdevicesinc/plutosdr-fw/).

2. Install dfu utility. On Ubuntu/Debian it's `apt install dfu-util`.

5. Make Device enter DFU mode. There is a howto section below on the details steps.

4. Run DFU utility with the following command:

  ```
  dfu-util -a 0 -D boot.dfu
  ```

dfu-util -a 1 -D pluto.dfu -R
```

The `-R` option is necessary to perform a USB Reset and make the device use the updated firmware. Depending on USB permissions given to your current user, you might need to prefix the commands with `sudo`.

# How to Enter DFU

1. Remove the screws and open up the device.
2. Short URST with MIO52, see the Jumper images for details.
3. Press DFU while power the device

https://guenniefoto.de/archive/20347
https://github.com/plutoplus/plutoplus/issues/51

https://github.com/plutoplus/plutoplus/issues/51

Set jumper, hold DFU
When booted sudo dfu-util -l
sudo dfu-util -a firmware.dfu -D pluto.dfu
sudo `dfu-util -a boot.dfu -D ./boot.dfu`
sudo `dfu-util -a uboot-env.dfu -D ./uboot-env.dfu`

`UNCLEAR WHY BUT newer timestamp 0.38 firmware does not work on some PLUTOPLUS`

Feb17
Setup bladerf1 pi5
sudo apt-get install git screen libiio-dev libiio-utils vim python3-dev uhubctl libusb-dev cmake
#sudo apt-get install libusb-1.0-0-dev # maybe dont need this one
sudo apt-get install libncurses5-dev

git clone --depth 1 https://github.com/Nuand/bladeRF.git
    git clone --depth 1 https://github.com/Nuand/bladeRF.git
    # cd bladeRF/host
    # mkdir build && cd build
    # cmake ..
    # make
    # sudo make install
```

sudo ldconfig # ** fixes libbladeRF.so.2 issue

USB permissions
sudo nano /etc/udev/rules.d/99-usb-permissions.rules
SUBSYSTEM=="usb", MODE="0666"
sudo udevadm control --reload-rules
sudo udevadm trigger

python -m venv ~/spf-virtualenv
source ~/spf-virtualenv/bin/activate
cd ~/bladeRF/host/libraries/libbladeRF_bindings/python/
python setup.py install
Cd ~/
Mkdir gits
Cd gits
git clone https://github.com/misko/spf.git
~/gits/spf $ pip install -e .

/boot/firmware/config.txt
dtoverlay=disable-wifi
dtoverlay=disable-bt

#------------------------------------ # Aggressive underclock on Pi 5: #------------------------------------
arm_freq=1200 # 1.2 GHz (down from ~2.4)
gpu_freq=300 # GPU at 300 MHz (normally ~500+)
gpu_mem=16 # minimal VRAM in MB if headless
over_voltage=-4 # negative offset to reduce voltage; test stability

echo "powersave" | sudo tee /sys/devices/system/cpu/cpufreq/policy0/scaling_governor

stress-ng --cpu 4 --timeout 60s # 0.85A
stress-ng --cpu 4 --timeout 60s # 1.0~0.85A Without underclocking
stress-ng --cpu 4 --timeout 60s # 0.9~0.80A With -4

Disabling HDMI didnt help much

sudo systemctl disable triggerhappy sudo systemctl disable avahi-daemon sudo systemctl disable hciuart

Install cpufrequtils
Modify /etc/init.d/cpufrequtils
sudo systemctl enable cpufrequtils sudo systemctl start cpufrequtils

Cpufreq utils not persistent

```
[Unit]
Description=Set CPU Governor at boot
After=multi-user.target

[Service]
Type=oneshot
ExecStart=/usr/bin/bash -c '\
   for cpu in /sys/devices/system/cpu/cpu[0-9]*; do \
      echo powersave > "$cpu/cpufreq/scaling_governor"; \
   done'
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl enable set-governor.service
sudo systemctl start set-governor.service

sudo vi /boot/firmware/cmdline.txt # maxcpus=2
```

Setup bladerf1 zero2

```
sudo apt-get install git screen libiio-dev libiio-utils vim python3-dev uhubctl libusb-dev
libusb-1.0-0-dev
sudo apt-get install bladerf
git clone --depth 1 https://github.com/Nuand/bladeRF.git
cd bladeRF/host/libraries/libbladeRF_bindings/python
```

ILLEGAL INSTRUCTION (in the end wasnt maybe related to bladerf?)

Did full compile of bladerf

```
sudo ln -s /usr/lib/aarch64-linux-gnu/libbladeRF.so.2 /usr/lib/aarch64-linux-gnu/libbladeRF.so

Pip install -e .
```

USB permissions

```
sudo nano /etc/udev/rules.d/99-usb-permissions.rules
SUBSYSTEM=="usb", MODE="0666"
sudo udevadm control --reload-rules
sudo udevadm trigger
```

Libbladerf.so.2 vs libbladerf.so , missing sudo apt-get install libbladerf-dev

python spf/sdrpluto/sdr_controller.py --receiver-uri ip:192.168.1.18 --mode rx --fs 16000000 --fi 10000000 --fc 5839000000 --rx-mode fast_attack --rx-n 131072 --rx-spacing 0.025

python spf/sdrpluto/sdr_controller.py --receiver-uri bladerf --mode rx --fs 32000000 --fi 10000000 --fc 5834000000 --rx-mode fast_attack --rx-n 131072 --rx-spacing 0.025

Feb 16

```
 2061  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission1 rover_2025_02_16_14_45_28_nRX2_diamond_spacing0p05075_tag_RO1
 2062  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission2 rover_2025_02_16_15_16_14_nRX2_diamond_spacing0p05075_tag_RO1
 2063  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission3 rover_2025_02_16_15_58_30_nRX2_diamond_spacing0p05075_tag_RO1
 2064  cd ../rover2/
 2065  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission1 rover_2025_02_16_14_45_04_nRX1_circle_spacing0p05075_tag_RO2
 2066  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission2 rover_2025_02_16_15_25_48_nRX1_circle_spacing0p05075_tag_RO2
 2067  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission3 rover_2025_02_16_15_58_20_nRX1_circle_spacing0p05075_tag_RO2
 2068  cd ../rover3/
 2069  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission1 rover_2025_02_16_14_45_43_nRX2_center_spacing0p05075_tag_RO3
 2070  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission2 rover_2025_02_16_15_22_16_nRX2_center_spacing0p05075_tag_RO3
 2071  bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb16_mission3 rover_2025_02_16_15_58_48_nRX2_center_spacing0p05075_tag_RO3
```

```
ms="1 2 3"; rovers="1 3"; for m in $ms; do for rover in $rovers; do echo python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/feb16_mission${m}.zarr --rx ../rover${rover}/feb16_mission${m}.zarr --output feb16_mission${m}_rover${rover}.zarr; done ; done
```

Setup bladerf1

```
git clone git@github.com:misko/spf.git
 python -m venv ../spf_venv
```

Feb 15

Debugging radios

New USB cables

TURN OF WIFI!!!
Make sure data on correct ports!!!
Flashed radio that was wonky ***

Nuand bladerf install setup
git clone --recursive https://github.com/Nuand/bladeRF.git
https://github.com/Nuand/bladeRF/tree/master/host
bladeRF-cli -v verbose -L ~/Downloads/hostedxA4-latest.rbf
sudo bladeRF-cli -e info -e version
Bladerf-tool info
Install soapySDR
https://github.com/pothosware/homebrew-pothos/wiki
brew install python-setuptools
brew install soapyplutosdr
How to run iio_info /Library/Frameworks/iio.framework/Tools/iio_info -S
https://github.com/pothosware/homebrew-pothos/wiki

cd host
Mkdir build
Cd build
sudo cmake -DTREAT_WARNINGS_AS_ERRORS=OFF ../
Sudo make
Sudo make install
Cd bladeRF/host/libraries/libbladeRF_bindings/python
Pip install setuptools
Python setup.py install

Binding DJI and activating air unit . to activate must first plug in aircraft power, then USB to computer

Feb11
python spf/scripts/create_empirical_p_dist.py -d /mnt/md2/cache/nosig_data/*.zarr --precompute-cache /mnt/md2/cache/precompute_cache_3p5/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_
Issues with rotation, ground truth craft theta was absolute theta, removed adding heading so now its just craft theta

Feb10
b2 sync --threads 16 /mnt/md2/ b2://projectspf/md2/
for x in /mnt/md2/2d_wallarray_v2_data/feb/*.zarr; do python ~/gits/spf/spf/scripts/zarr_rechunk.py --no-check-contents -s $x /mnt/md2/cache/nosig_data/`basename $x`; done

ls -d /mnt/md2/2d_wallarray_v2_data/feb/*.zarr | shuf | while read x ; do python ~/gits/spf/spf/scripts/segment_zarr.py -i $x --precompute-cache /mnt/md2/cache/precompute_cache_3p5/ -w 0 -p 16; done

Feb 9
Add swap
sudo fallocate -l 256G /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile

Feb 9
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/feb9_mission1.zarr --rx ../rover1/feb9_mission1.zarr --output feb9_mission1_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/feb9_mission1.zarr --rx ../rover3/feb9_mission1.zarr --output feb9_mission1_rover3.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/feb9_mission2.zarr --rx ../rover1/feb9_mission2.zarr --output feb9_mission2_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/feb9_mission2.zarr --rx ../rover3/feb9_mission2.zarr --output feb9_mission2_rover3.zarr

Feb 7
 2023 bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb7_mission1 rover_2025_02_07_14_48_36_nRX2_diamond_spacing0p05075_tag_RO1
 2025 bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb7_mission1 rover_2025_02_07_14_54_37_nRX1_circle_spacing0p05075_tag_RO2
 2029 bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb7_mission1 rover_2025_02_07_14_48_47_nRX2_center_spacing0p05075_tag_RO3

python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/feb7_mission1.zarr --rx ../rover1/feb7_mission1.zarr --output feb7_mission1_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/feb7_mission1.zarr --rx ../rover3/feb7_mission1.zarr --output feb7_mission1_rover3.zarr

Feb 6
Signal is weakish
	Things to try raise emitter?
How to get to 5.8ghz?

Feb 5

 2011 bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb5_mission1 rover_2025_02_05_15_01_34_nRX2_diamond_spacing0p05075_tag_RO1

2015 bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb5_mission1 rover_2025_02_05_15_01_07_nRX1_circle_spacing0p05075_tag_RO2
2020 bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh feb5_mission1 rover_2025_02_05_15_01_23_nRX2_center_spacing0p05075_tag_RO3

python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/feb5_mission1.zarr --rx ../rover1/feb5_mission1.zarr --output feb5_mission1_rover1.zarr

RESET cache
sudo dmsetup remove md2_cached
sudo dmsetup remove cache_data
sudo dmsetup remove cache_meta
sudo dd if=/dev/zero of=/dev/md126 bs=1M count=16

512kb ** using this one
sudo dmsetup create cache_meta --table "0 1433600 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46874566144 linear /dev/md126 1433600"
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 1024 1 writethrough default 0"
sudo dmsetup message md2_cached 0 "migration_threshold 0"

Feb 2
Ad9361 ad9363 ad9364
https://wiki.analog.com/resources/eval/user-guides/ad-fmcomms2-ebz/introduction

Feb 1
Updated v5 notebook
Issues identified with rover gain settings
Installing https://wiki.analog.com/resources/tools-software/linux-software/iio_oscilloscope

HOMEBREW_NO_INSTALL_FROM_API=1 brew install --interactive --build-from-source ./iio-oscilloscope.rb
Running on parallels desktop, 0.15?

What is filter?
https://ez.analog.com/adieducation/university-program/f/q-a/104179/ad9361-fir-filter-configuration-with-matlab

iio_info -u ip:192.168.1.17 | grep -i agc

iio_info -u ip:192.168.1.17 | less

miskodzamba@Miskos-MBP ~ % iio_attr -u ip:192.168.1.18 -c ad9361-phy voltage0
dev 'ad9361-phy', channel 'voltage0' (input), attr 'bb_dc_offset_tracking_en', value '1'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'filter_fir_en', value '1'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'gain_control_mode', value 'fast_attack'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'gain_control_mode_available', value 'manual fast_attack slow_attack hybrid'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'hardwaregain', value '19.000000 dB'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'hardwaregain_available', value '[-3 1 71]'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'quadrature_tracking_en', value '1'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'rf_bandwidth', value '300000'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'rf_bandwidth_available', value '[200000 1 56000000]'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'rf_dc_offset_tracking_en', value '1'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'rf_port_select', value 'A_BALANCED'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'rf_port_select_available', value 'A_BALANCED B_BALANCED C_BALANCED A_N A_P B_N B_P C_N C_P TX_MONITOR1 TX_MONITOR2 TX_MONITOR1_2'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'rssi', value '45.00 dB'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'sampling_frequency', value '30000000'
dev 'ad9361-phy', channel 'voltage0' (input), attr 'sampling_frequency_available', value '[1041666 1 30720000]'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'filter_fir_en', value '1'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'hardwaregain', value '-80.000000 dB'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'hardwaregain_available', value '[-89.750000]'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'rf_bandwidth', value '18000000'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'rf_bandwidth_available', value '[200000 1 40000000]'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'rf_port_select', value 'A'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'rf_port_select_available', value 'A B'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'rssi', value '0.00 dB'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'sampling_frequency', value '30000000'
dev 'ad9361-phy', channel 'voltage0' (output), attr 'sampling_frequency_available', value '[1041666 1 30720000]'
miskodzamba@Miskos-MBP ~ % iio_attr -u ip:192.168.1.18 -c ad9361-phy voltage0 filter_fir_en

https://ez.analog.com/adieducation/university-program/f/q-a/104179/ad9361-fir-filter-configuration-with-matlab

January 30

python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/jan30_mission1.zarr --rx ../rover3/jan30_mission1.zarr --output jan30_mission1_rover3.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/jan30_mission2.zarr --rx ../rover3/jan30_mission2.zarr --output jan30_mission2_rover3.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/jan30_mission1.zarr --rx ../rover1/jan30_mission1.zarr --output jan30_mission1_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/jan30_mission2.zarr --rx ../rover1/jan30_mission2.zarr --output jan30_mission2_rover1.zarr

January 21
Test drone (docker)
python spf/mavlink_radio_collection.py --yaml-config tests/rover_config.yaml --device-mapping tests/test_device_mapping -r circle -n 1000 --temp /tmp/ --exit

January 15
RESET cache
sudo dmsetup remove md2_cached
sudo dmsetup remove cache_data
sudo dmsetup remove cache_meta
sudo dd if=/dev/zero of=/dev/md126 bs=1M count=16

512kb ** using this one
sudo dmsetup create cache_meta --table "0 1433600 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46874566144 linear /dev/md126 1433600"
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 1024 1 writethrough default 0"
sudo dmsetup message md2_cached 0 "migration_threshold 0"

128kb
sudo dmsetup create cache_meta --table "0 5734400 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46875053184 linear /dev/md126 5734400"
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 256 1 writethrough default 0"
sudo dmsetup message md2_cached 0 "migration_threshold 0"

January 13
b2 sync --threads 16 /mnt/md2/2d_wallarray_v2_data b2://projectspf/datasets/full_data/
while [1 -lt 2]; do rsync -avP --remove-source-files pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.zarr' ./ ; rsync -avP

pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.yaml' ./ ; sleep 600; done

January 5

1.2 Two Types of DC Offset Calibration in AD9361

1. **RF DC Offset Correction (RFDC)**
    - Occurs in the **RF front-end** to remove offset introduced around the mixer and LNA stages.
    - Primarily combats LO leakage or imbalances around the RF section.
    - Also referred to as **RF DC Offset Tracking**.
2. **Baseband DC Offset Correction (BBDC)**
    - Occurs in the **baseband** section (filters, amplifiers, ADC stages).
    - Removes offsets introduced after downconversion, ensuring a clean baseband signal.
    - Also called **BB DC Offset Tracking**.

January 4
https://wiki.analog.com/university/tools/pluto/drivers/osx
OSX pluto plus

January 2
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/591066/iq-centering-dc-offset-not-working-as-expected
Same issue here?
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/80231/baseband-dc-offset-problem-on-ad9361

https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/586259/bb-dc-offset-tracker
https://chatgpt.com/c/67749ca8-3ed8-8002-9877-160dfdbccdf5 [the issue might be that the IF is 0hz? And part of the DC component? This leads to weird behavior

Disable unused port
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/589604/disable-unuse-tx-and-rx-port

Cool docs

https://gigayasawireless.github.io/5G-on-SDRs/index.html

Sampling rate
https://ez.analog.com/adieducation/university-program/f/q-a/165546/pluto-sdr-buffer-in-gnu-radio

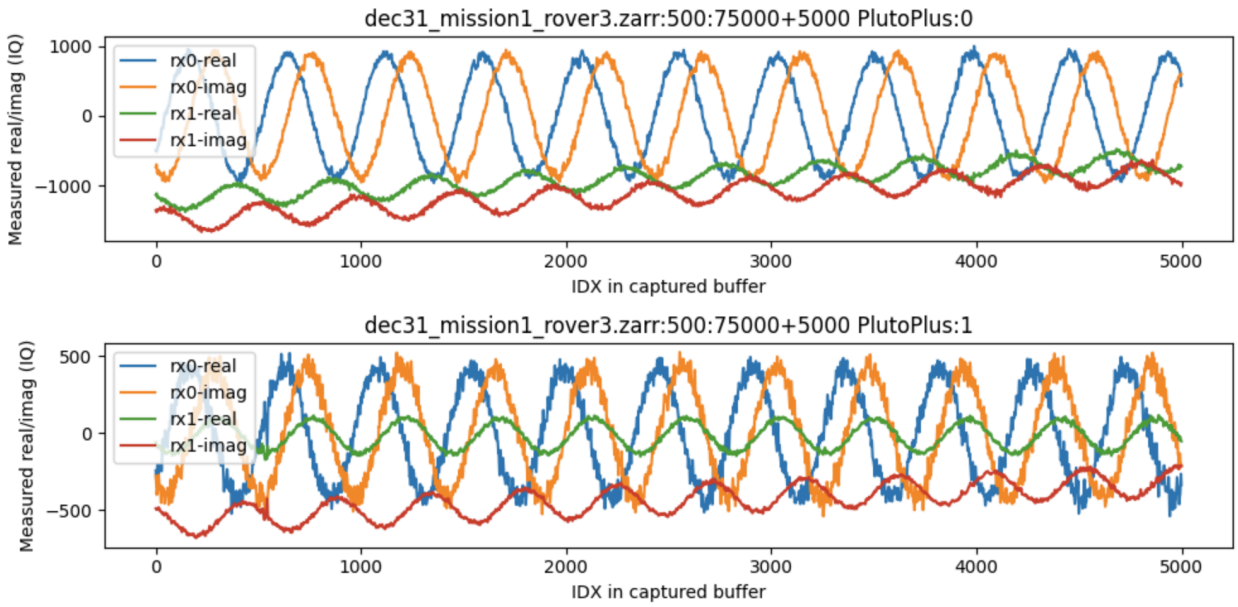
January 1 2025
IQ correction
https://wiki.analog.com/resources/eval/user-guides/ad-fmcomms1-ebz/iq_correction

Get plutoplus revision
cat /proc/device-tree/model

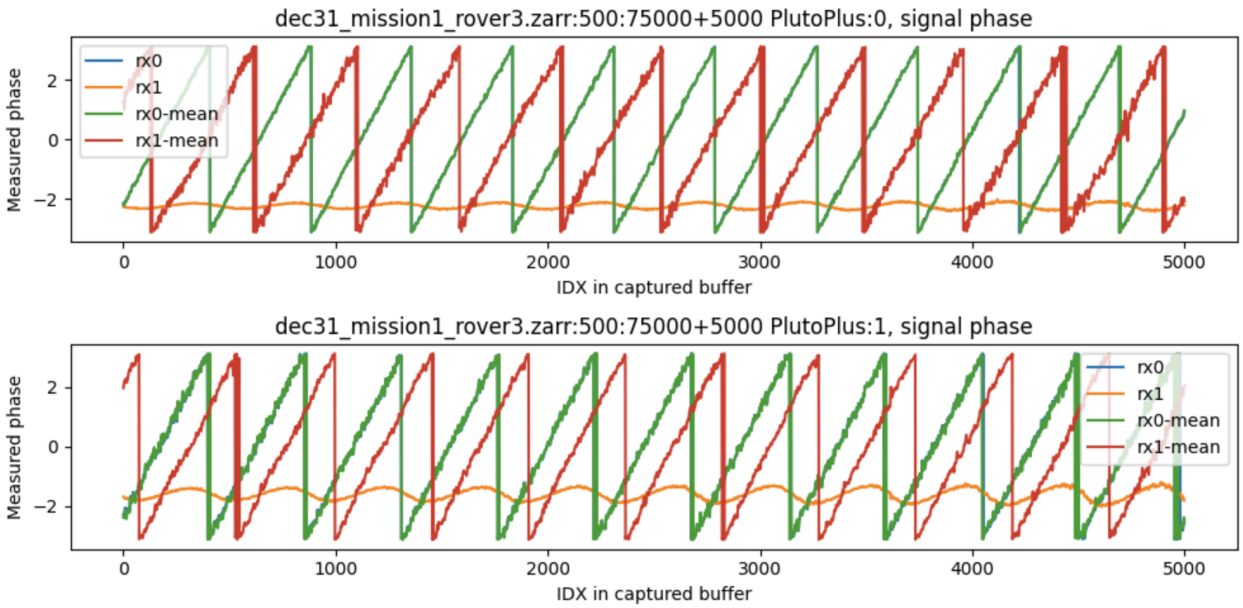
AD9361 customization
https://wiki.analog.com/resources/tools-software/linux-drivers/iio-transceiver/ad9361-customization

Device tree check
find . | grep 2r
./amba/spi@e0006000/ad9361-phy@0/adi,2rx-2tx-mode-enable
cat ./amba/spi@e0006000/ad9361-phy@0/adi,2rx-2tx-mode-enable
find . | grep 2r
./amba/spi@e0006000/ad9361-phy@0/adi,2rx-2tx-mode-enable
pwd
/sys/firmware/devicetree/base

DC offset correction is an issue!! On RX1!



1. **IQ Imbalance Calibration**
2. **DC Offset Correction**
3. **IQ Centering**
4. **Automatic Gain Control (AGC)**
5. **Image Rejection**
6. **Frequency Offset Correction**
7. **AD9361 Calibration**
8. **RF Front-End Calibration**
9. **IQ Crosstalk**
10. **Quadrature Mismatch**
11. **Phase Noise**
12. **DC Cancellation**
13. **LO Leakage**
14. **Baseband Correction**
15. **IQ Data Alignment**



Can detrend window, and seem to get good results!

December 31
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec31_mission1
rover_2024_12_31_15_06_21_nRX2_center_spacing0p05075_tag_RO1

Gain issues,
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/80250/ad9361-rx1-rx2-polarity-inversion

Linux API ad9361
https://wiki.analog.com/resources/tools-software/linux-drivers/iio-transceiver/ad9361?s

December 30
Controller setup ACCESS
Setup new model
Setup -> external RF -> R9M Access
Set RXNUM unique to rover
Press Reg
***Power on receiver while holding the one weird button
***
Enter / Register on controller
*** Power off receiver

***
Press bind on controller
*** Power on receiver
Bind it
Press receiver1 -> options -> CH6 = SBUS

Inputs
THR
AIL
ELE
RUD

Mixes
1 Rud
2 Ele
3 Thr
4 Ail
5 SF
6 S2
7 SD
8 SA
9 SH
10 SC
11 S2
12 SB

December 29
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/112556/ad9361-is-the-full-gain-table-shared-between-the-two-rx-channels

Building pluto plus FW
sudo apt install bison
sudo apt install flex

```
git clone --branch v0.38_plutoplus_timestamp --recurse-submodules
--shallow-submodules https://github.com/pgreenland/plutosdr-fw.git
cd plutosdr-fw
make
```

CROSS COMPILE GCC

export PATH=/home/mouse9911/gits/plutoplus/plutosdr-fw/buildroot/output/host/bin:$PATH
export CROSS_COMPILE=arm-buildroot-linux-gnueabihf-

https://github.com/analogdevicesinc/buildroot/issues/65
ERROR: ad936x_ref_cal-01747db5cd60ff64115a73ac1f3bb97911f5c58e-br1.tar.gz has wrong sha256 hash:
ERROR: expected:
26aedd8021fa939ab2f53e55904d869207265242fef7ad86aa4673e219b7cbef
ERROR: got : 4814915de63d975807e918df82bb86021d0e78839e8cc4116a36476d0b33180c
ERROR: Incomplete download, or man-in-the-middle (MITM) attack

PLUTO BUILDROOT DIFF

```
diff --git a/package/ad936x_ref_cal/ad936x_ref_cal.hash
b/package/ad936x_ref_cal/ad936x_ref_cal.hash
index 24eb7fb8..4ab5a059 100644
--- a/package/ad936x_ref_cal/ad936x_ref_cal.hash
+++ b/package/ad936x_ref_cal/ad936x_ref_cal.hash
@@ -1,5 +1,6 @@
 # Locally computed
-sha256 26aedd8021fa939ab2f53e55904d869207265242fef7ad86aa4673e219b7cbef
ad936x_ref_cal-01747db5cd60ff64115a73ac1f3bb97911f5c58e-br1.tar.gz
+#sha256 26aedd8021fa939ab2f53e55904d869207265242fef7ad86aa4673e219b7cbef
ad936x_ref_cal-01747db5cd60ff64115a73ac1f3bb97911f5c58e-br1.tar.gz
+sha256 4814915de63d975807e918df82bb86021d0e78839e8cc4116a36476d0b33180c
ad936x_ref_cal-01747db5cd60ff64115a73ac1f3bb97911f5c58e-br1.tar.gz

 # License files (locally computed as well)
 sha256 8177f97513213526df2cf6184d8ff986c675afb514d4e68a404010521b880643 LICENSE
diff --git a/package/libiio/libiio.hash b/package/libiio/libiio.hash
index f1c58efb..48dea497 100644
--- a/package/libiio/libiio.hash
+++ b/package/libiio/libiio.hash
@@ -1,6 +1,7 @@
 # Locally calculated
 sha256 a2b5d848531ea64fd9f95327dfd5a588bd227d9577281ec375e822702c6a52d5
libiio-0.24.tar.gz
 sha256 21972599a3c143ab1f98002ad2b3f28f4aff927fde5f677478311cd4e517730c
libiio-0.25.tar.gz
-sha256 e791ad1cf35aef08fc6e2b6b0dcdd1cc21d36cf287d81fa14adb088c6c1d4c49
libiio-38483f31be391af66b35542f733e569febe13d3a-br1.tar.gz
+#sha256 e791ad1cf35aef08fc6e2b6b0dcdd1cc21d36cf287d81fa14adb088c6c1d4c49
libiio-38483f31be391af66b35542f733e569febe13d3a-br1.tar.gz
```

+sha256 865fe496624bebc7c4266bf157fb4b5ec851b5f929d2571ec73846e4f317cf46
libiio-38483f31be391af66b35542f733e569febe13d3a-br1.tar.gz
 sha256 102900208eef27b766380135906d431dba87edaa7ec6aa72e6ebd3dd67f3a97b
COPYING.txt

PLUTO_FW DIFF

```
diff --git a/Makefile b/Makefile
index 0dc9c4b..cbad6f9 100644
--- a/Makefile
+++ b/Makefile
@@ -14,7 +14,8 @@ VERSION=$(shell git describe --abbrev=4 --dirty --always --tags)
 LATEST_TAG=$(shell git describe --abbrev=0 --tags)
 UBOOT_VERSION=$(shell echo -n "PlutoSDR " && cd u-boot-xlnx && git describe --abbrev=0
--dirty --always --tags)
 HAVE_VIVADO= $(shell bash -c "source $(VIVADO_SETTINGS) > /dev/null 2>&1 && vivado
-version > /dev/null 2>&1 && echo 1 || echo 0")
-XSA_URL ?=
http://github.com/analogdevicesinc/plutosdr-fw/releases/download/${LATEST_TAG}/system_top.
xsa
+#XSA_URL ?=
http://github.com/analogdevicesinc/plutosdr-fw/releases/download/${LATEST_TAG}/system_top.
xsa
+XSA_URL ?=
http://github.com/pgreenland/plutosdr-fw/releases/download/${LATEST_TAG}/system_top.xsa

 ifeq (1, ${HAVE_VIVADO})
        VIVADO_INSTALL= $(shell bash -c "source $(VIVADO_SETTINGS) > /dev/null 2>&1 &&
vivado -version | head -1 | awk '{print $2}'")
diff --git a/buildroot b/buildroot
--- a/buildroot
+++ b/buildroot
@@ -1 +1 @@
-Subproject commit af0a100f5479ccbf46a8ca0cfcb3907e87f3e9ce
+Subproject commit af0a100f5479ccbf46a8ca0cfcb3907e87f3e9ce-dirty
```

December 28

```
#merge data
rovers="3 1"; missions="1 2 3 4"; for rover in $rovers; do for mission in $missions; do echo
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec28_mission${mission}.zarr --rx
```

../rover${rover}/dec28_mission${mission}.zarr --output
dec28_mission${mission}_rover${rover}.zarr; done; done

December 28 bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission1
rover_2024_12_28_16_08_24_nRX1_circle_spacing0p05075_tag_RO2.zarr
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission2
rover_2024_12_28_16_35_10_nRX1_circle_spacing0p05075_tag_RO2.
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission3
rover_2024_12_28_17_00_41_nRX1_circle_spacing0p05075_tag_RO2.
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission4
rover_2024_12_28_17_25_30_nRX1_circle_spacing0p05075_tag_RO2.

bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission1
rover_2024_12_28_16_09_49_nRX2_center_spacing0p05075_tag_RO1
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission2
rover_2024_12_28_16_35_09_nRX2_center_spacing0p05075_tag_RO1
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission3
rover_2024_12_28_17_00_16_nRX2_center_spacing0p05075_tag_RO1
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission4
rover_2024_12_28_17_25_24_nRX2_center_spacing0p05075_tag_RO1

bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission1
rover_2024_12_28_16_07_41_nRX2_bounce_spacing0p05075_tag_RO3
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission2
rover_2024_12_28_16_34_56_nRX2_bounce_spacing0p05075_tag_RO3
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission3
rover_2024_12_28_17_00_05_nRX2_bounce_spacing0p05075_tag_RO3
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec28_mission4
rover_2024_12_28_17_25_18_nRX2_bounce_spacing0p05075_tag_RO3

December 27

python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec27_mission1.zarr --rx
../rover1/dec27_mission1.zarr --output dec27_mission1_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec27_mission2.zarr --rx
../rover1/dec27_mission2.zarr --output dec27_mission2_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec27_mission3.zarr --rx
../rover1/dec27_mission3.zarr --output dec27_mission3_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec27_mission3.zarr --rx
../rover3/dec27_mission3.zarr --output dec27_mission3_rover3.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec27_mission2.zarr --rx
../rover3/dec27_mission2.zarr --output dec27_mission2_rover3.zarr

python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec27_mission1.zarr --rx ../rover3/dec27_mission1.zarr --output dec27_mission1_rover3.zarr

bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec27_mission1 rover_2024_12_27_15_57_02_nRX1_circle_spacing0p05075_tag_RO2
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec27_mission2 rover_2024_12_27_16_22_00_nRX1_circle_spacing0p05075_tag_RO2
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec27_mission3 rover_2024_12_27_16_51_15_nRX1_circle_spacing0p05075_tag_RO2

bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec27_mission1 rover_2024_12_27_15_52_20_nRX2_center_spacing0p05075_tag_RO1
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec27_mission2 rover_2024_12_27_16_22_19_nRX2_center_spacing0p05075_tag_RO1
bash /home/mouse9911/gits/spf/spf/scripts/v4_link.sh dec27_mission3 rover_2024_12_27_16_48_15_nRX2_center_spacing0p05075_tag_RO1

mv rover_2024_12_27_16_48_31_nRX2_bounce_spacing0p05075_tag_RO3.log.tmp rover_2024_12_27_16_48_31_nRX2_bounce_spacing0p05075_tag_RO3.log
mv rover_2024_12_27_16_48_31_nRX2_bounce_spacing0p05075_tag_RO3.yaml.tmp rover_2024_12_27_16_48_31_nRX2_bounce_spacing0p05075_tag_RO3.yaml
mv rover_2024_12_27_16_48_31_nRX2_bounce_spacing0p05075_tag_RO3.zarr.tmp rover_2024_12_27_16_48_31_nRX2_bounce_spacing0p05075_tag_RO3.zarr

December 26

Create empirical
python spf/scripts/create_empirical_p_dist.py -d /mnt/ssd/cache/nosig_data/*.zarr --precompute-cache /mnt/ssd/cache/precompute_cache_3p4/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

Create nosigs
for x in /mnt/md2/2d_wallarray_v2_data/*/*.zarr; do python ~/gits/spf/spf/scripts/zarr_rechunk.py --no-check-contents -s $x /mnt/md2/cache/nosig_data/`basename $x`; done

bash ~/gits/spf/spf/scripts/v4_link.sh dec26_mission1 rover_2024_12_26_20_42_17_nRX1_circle_spacing0p05075_tag_RO2
bash ~/gits/spf/spf/scripts/v4_link.sh dec26_mission2 rover_2024_12_26_21_12_15_nRX1_circle_spacing0p05075_tag_RO2

bash ~/gits/spf/spf/scripts/v4_link.sh dec26_mission1 rover_2024_12_26_20_40_42_nRX2_center_spacing0p05075_tag_RO1
bash ~/gits/spf/spf/scripts/v4_link.sh dec26_mission2 rover_2024_12_26_21_12_01_nRX2_bounce_spacing0p05075_tag_RO3

#merge
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec26_mission1.zarr --rx ../rover1/dec26_mission1.zarr --output dec26_mission1_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec26_mission2.zarr --rx ../rover1/dec26_mission2.zarr --output dec26_mission2_rover1.zarr

December 25
S3
aws configure set role_arn arn:aws:iam::715841335207:user/mouse9911
Created user iam
Created access keys
Ran
Aws configure (on kalman)
https://docs.aws.amazon.com/cli/latest/userguide/cli-authentication-user.html#cli-authentication-user-get
https://docs.aws.amazon.com/cli/latest/userguide/getting-started-quickstart.html

Rechunk dataset to nosig
for x in /mnt/md2/2d_wallarray_v2_data/*/*.zarr; do python ~/gits/spf/spf/scripts/zarr_rechunk.py --no-check-contents -s $x /mnt/md2/cache/nosig_data/`basename $x`; done

Setting up backblaze for storage
Wget https://github.com/Backblaze/B2_Command_Line_Tool/releases/latest/download/b2-linux
./b2-linux ls b2://projectspf/
b2 account authorize

bash ~/gits/spf/spf/scripts/v4_link.sh dec25_mission1 rover_2024_12_25_15_31_17_nRX2_bounce_spacing0p05075_tag_RO3
bash ~/gits/spf/spf/scripts/v4_link.sh dec25_mission1 rover_2024_12_25_15_31_44_nRX2_center_spacing0p05075_tag_RO1
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec25_mission1.zarr --rx ../rover1/dec25_mission1.zarr --output dec25_mission1_rover1.zarr
python ~/gits/spf/spf/scripts/v4_tx_rx_to_v5.py --tx ../rover2/dec25_mission1.zarr --rx ../rover3/dec25_mission1.zarr --output dec25_mission1_rover3.zarr

December 23
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/dec23_mission1.zarr --rx ../rover1/dec23_mission1.zarr --output dec23_mission1_rover1.zarr
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/dec23_mission1.zarr --rx ../rover3/dec23_mission1.zarr --output dec23_mission1_rover3.zarr

Split val and train by bandwidth

December 20

Flashed rover 2 with tone blaster

```
Serial1.println("wifiscwout 1 12 1"); // channel 12
```

December 19
ls /mnt/ssd/2d_wallarray_v2_data/*/*.zarr -d > /tmp/dec19_full.txt

ls /mnt/ssd/2d_wallarray_v2_data/*/*.zarr -d > /mnt/md2/splits/dec19_full.txt

Create empirical distribution
python spf/scripts/create_empirical_p_dist.py -d /mnt/ssd/cache/nosig_data/*.zarr --precompute-cache /mnt/ssd/cache/precompute_cache_3p4/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

December 18

Create summaries for wandb
Start paired training
Kick off baseline runs for new data
Merge rover runs

python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/dec18_mission1.zarr --rx ../rover1/dec18_mission1.zarr --output dec18_mission1_rover1.zarr
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/dec18_mission1.zarr --rx ../rover3/dec18_mission1.zarr --output dec18_mission1_rover3.zarr
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/dec18_mission2.zarr --rx ../rover1/dec18_mission2.zarr --output dec18_mission2_rover1.zarr
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/dec18_mission2.zarr --rx ../rover3/dec18_mission2.zarr --output dec18_mission2_rover3.zarr

December 15

PLUTO SETUP
https://github.com/misko/spf/blob/main/spf/sdrpluto/hardware_setup/setup

Config.txt ethernet usb_ethernet usb rndis

Nosig

for x in /mnt/md2/2d_wallarray_v2_data/*/*.zarr; do echo python spf/dataset/zarr_rechunk.py --no-check-contents -s $x /mnt/md2/cache/nosig_data/`basename $x`; done

Decembet 8

```
cat /mnt/md2/splits/dec7_train.txt | while read line; do grep 2412 ${line}.ls.json > /dev/null; if [ $? -eq 1 ] ; then echo $line; fi ; done > /mnt/md2/splits/dec7_train_no24.txt
cat /mnt/md2/splits/dec7_val.txt | while read line; do grep 2412 ${line}.ls.json > /dev/null; if [ $? -eq 1 ] ; then echo $line; fi ; done > /mnt/md2/splits/dec7_val_no24.txt

cat /mnt/md2/splits/dec7_train.txt | while read line; do grep 2412 ${line}.ls.json > /dev/null; if [ $? -eq 0 ] ; then echo $line; fi ; done > /mnt/md2/splits/dec7_train_24.txt
cat /mnt/md2/splits/dec7_val.txt | while read line; do grep 2412 ${line}.ls.json > /dev/null; if [ $? -eq 0 ] ; then echo $line; fi ; done > /mnt/md2/splits/dec7_val_24.txt
```

December 7

FINAL FINAL FINAL
https://blog.kylemanna.com/linux/ssd-caching-using-dmcache-tutorial/

```
sudo dmsetup remove md2_cached
sudo dmsetup remove cache_data
sudo dmsetup remove cache_meta
sudo dd if=/dev/zero of=/dev/md126 bs=1M count=16
sudo blockdev --getss /dev/md126
sudo blockdev --getss /dev/md2
sudo blockdev --getsize64 /dev/md
sudo blockdev --getsize64 /dev/md126
~~sudo dmsetup create cache_meta --table "0 1048576 linear /dev/md126 0"~~
~~sudo dmsetup create cache_data --table "0 46876735488 linear /dev/md126 1048576"~~
sudo dmsetup create cache_meta --table "0 262144 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46875736192 linear /dev/md126 262144"
sudo dmsetup table cache_meta
sudo dmsetup table cache_data
~~sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 2048 1 writethrough mq 0"~~
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 16384 1 writethrough mq 0"
sudo dmsetup message md2_cached 0 "migration_threshold 0"

sudo dmsetup create cache_meta --table "0 262144 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46875736192 linear /dev/md126 262144"
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 16384 1 writethrough mq 0"
```

sudo mount /dev/mapper/md2_cached /mnt/md2

sudo dmsetup create cache_meta --table "0 1433600 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46874566144 linear /dev/md126 1433600"
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta
/dev/mapper/cache_data /dev/md2 1024 1 writethrough mq 0"

sudo dmsetup create cache_meta --table "0 1433600 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46874566144 linear /dev/md126 1433600"
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta
/dev/mapper/cache_data /dev/md2 512 1 writethrough mq 0"

sudo dmsetup create cache_meta --table "0 1433600 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46874566144 linear /dev/md126 1433600"
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta
/dev/mapper/cache_data /dev/md2 1024 1 writethrough default 0"
sudo dmsetup message md2_cached 0 "migration_threshold 0"

128kb
sudo dmsetup create cache_meta --table "0 5734400 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 46875053184 linear /dev/md126 5734400"
sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta
/dev/mapper/cache_data /dev/md2 256 1 writethrough default 0"
sudo dmsetup message md2_cached 0 "migration_threshold 0"

Final
mouse9911@kalman:~/gits/spf$ sudo dmsetup remove cache_meta
sudo dmsetup remove cache_data
sudo dmsetup remove md2_cached
device-mapper: remove ioctl on md2_cached failed: No such device or address
Command failed.
mouse9911@kalman:~/gits/spf$ sudo dmsetup ls
ubuntu--vg-ubuntu--lv   (252:0)
mouse9911@kalman:~/gits/spf$ sudo dd if=/dev/zero of=/dev/md126 bs=1M count=16
16+0 records in
16+0 records out
16777216 bytes (17 MB, 16 MiB) copied, 0.00909684 s, 1.8 GB/s

```
mouse9911@kalman:~/gits/spf$ sudo dmsetup create cache_meta --table "0 3145728 linear
/dev/md126 0"
mouse9911@kalman:~/gits/spf$ sudo dmsetup create cache_data --table "0 5856230400 linear
/dev/md126 3145728"
mouse9911@kalman:~/gits/spf$ sudo dmsetup create md2_cached --table "0 117190158336
cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 1024 1 writethrough default
0"

iostat -dx 5 /dev/mapper/md2_cached /dev/md2 /dev/md126

FINAL v2
mouse9911@kalman:~/gits/spf$ sudo dmsetup remove cache_data
mouse9911@kalman:~/gits/spf$ sudo dmsetup remove cache_meta
mouse9911@kalman:~/gits/spf$ sudo dd if=/dev/zero of=/dev/md126 bs=1M count=16
16+0 records in
16+0 records out
16777216 bytes (17 MB, 16 MiB) copied, 0.00935544 s, 1.8 GB/s
mouse9911@kalman:~/gits/spf$ sudo dmsetup create cache_meta --table "0 3145728 linear
/dev/md126 0"
mouse9911@kalman:~/gits/spf$ sudo dmsetup create cache_data --table "0 5856230400 linear
/dev/md126 3145728"
mouse9911@kalman:~/gits/spf$ sudo dmsetup create md2_cached --table "0 117190158336
cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 4096 1 writethrough default
0"

Finalv3
mouse9911@kalman:~/gits/spf$ sudo dmsetup remove md2_cached
mouse9911@kalman:~/gits/spf$ sudo dmsetup remove cache_meta
mouse9911@kalman:~/gits/spf$ sudo dmsetup remove cache_data
mouse9911@kalman:~/gits/spf$ sudo dd if=/dev/zero of=/dev/md126 bs=1M count=16
16+0 records in
16+0 records out
16777216 bytes (17 MB, 16 MiB) copied, 0.00858081 s, 2.0 GB/s
mouse9911@kalman:~/gits/spf$ sudo dmsetup create cache_meta --table "0 3145728 linear
/dev/md126 0"
mouse9911@kalman:~/gits/spf$ sudo dmsetup create cache_data --table "0 5856230400 linear
/dev/md126 3145728"
mouse9911@kalman:~/gits/spf$ sudo dmsetup create md2_cached --table "0 117190158336
cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 2048 1 writethrough default
0"

Restore dm-cache ssd nvme
/usr/local/bin/restore-dm-cache.sh
sudo dmsetup message md2_cached 0 "migration_threshold 2"
```

Faster promote?

sudo dmsetup message md2_cached 0 "flush"
sudo dmsetup remove md2_cached

sudo dmsetup remove md2_cached
sudo dmsetup remove cache_meta
sudo dmsetup remove cache_data
sudo dd if=/dev/zero of=/dev/md126 bs=1M count=16
sudo dmsetup create cache_meta --table "0 3145728 linear /dev/md126 0"
sudo dmsetup create cache_data --table "0 5856230400 linear /dev/md126 3145728"

sudo dmsetup create md2_cached --table "0 117190158336 cache /dev/mapper/cache_meta /dev/mapper/cache_data /dev/md2 2048 1 writethrough mq 0"
sudo dmsetup message md2_cached 0 "migration_threshold 0"

https://git.tnonline.net/Forza/dm-cache/src/branch/main

https://www.digitalocean.com/community/tutorials/how-to-use-lvm-to-manage-storage-devices-on-ubuntu-18-04

Calculating dm-cache metadata size
https://blog.kylemanna.com/linux/ssd-caching-using-dmcache-tutorial/
24004682317824*16/`(2**18)`

https://chatgpt.com/share/6754e76a-9efc-8002-bcd9-7da58b9f2ffc

Create physical volume from raid
sudo pvcreate /dev/md126
Create volume group
sudo vgcreate nvme_raid_vg /dev/md126

https://blog.delouw.ch/2020/01/29/using-lvm-cache-for-storage-tiering/
https://ahelpme.com/linux/lvm/ssd-cache-device-to-a-software-raid5-using-lvm2/#google_vignette
https://wiki.tnonline.net/w/Blog/dm-cache:_Linux_Accelerated_Storage

https://www.kernel.org/doc/html/v5.18/driver-api/md/raid5-cache.html
https://cristian.seceleanu.co.uk/2020/09/mdadm-caching/

December 4
Setting CPU performance governor
https://chatgpt.com/share/67510e32-5b24-8002-96f8-cbc3bf0ac786

Set fan speed
sudo liquidctl --match Elite set "fan" speed 100
sudo liquidctl --match Elite set "fan1" speed 100

Create LS
python spf/scripts/ls_ds.py -i /mnt/md0/spf/2d_wallarray_v2_data/*/*.zarr /mnt/md1/2d_wallarray_v2_data/*/*.zarr > ls_ds_dec1.csv

WandB report
https://wandb.ai/projectspf/2024_nov22_single_paired_multi/reports/Segmentation-3p11-vs-3p4--VmlldzoxMDQ2MTE5OA
Segmentation 3p4 3p11 3p3 windowed

xs="dec3_single_config_sigma0p05_rerun_3p4_windows_beamformerwindow_nophase_nobeam.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_beamformerwindow.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_norm.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_nlayers4.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_hidden64.yaml"; for x in $xs; do python spf/scripts/train_single_point.py --config latest_configs/$x; done

December 3
Renice priority for training
while [1 -lt 2]; do ps -aux | grep python | grep -v segment | grep train_singl | awk '{print $2}' | while read line; do renice -n -10 -p $line; done; sleep 300; done

xs="dec3_single_config_sigma0p05_rerun_3p4.yaml dec3_single_config_sigma0p05_rerun_3p4_windows.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_nobeam.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_nophase.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_norm.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_nlayers4.yaml dec3_single_config_sigma0p05_rerun_3p4_windows_hidden64.yaml"; for x in $xs; do python spf/scripts/train_single_point.py --config latest_configs/$x; done

December 2
mouse9911@kalman:~/gits/spf/latest_configs$ cat /mnt/4tb_ssd/nosig_data/train_fullpath.txt | while read line; do grep `basename $line` /tmp/full_md2 | head -n 1; done > /mnt/4tb_ssd/nosig_data/train_fullpath_md2.txt

mouse9911@kalman:~/gits/spf/latest_configs$ cat /mnt/4tb_ssd/nosig_data/val_fullpath.txt | while read line; do grep `basename $line` /tmp/full_md2 | head -n 1; done > /mnt/4tb_ssd/nosig_data/val_fullpath_md2.txt

Debug raid
sudo mdadm --examine /dev/sd*
sudo mdadm --assemble --scan

November 30
Mdadm skip raid check idle
echo idle | sudo tee /sys/block/md0/md/sync_action

Segment 3.4
python spf/scripts/segment_zarr.py -i `cat  /mnt/4tb_ssd/nosig_data/train_val_fullpath.txt` --precompute-cache /mnt/4tb_ssd/precompute_cache_3p4/

Fix up precomputed values
python spf/dataset//precompute_3p3_to_3p31.py -y /mnt/4tb_ssd/precompute_cache_3p31/wallarrayv3_2024_*.yarr

Compare git commits find the bugn
https://github.com/misko/spf/compare/e115701...ab184a0

cat /mnt/4tb_ssd/nosig_data/val.txt /mnt/4tb_ssd/nosig_data/train.txt | while read fn; do z=`basename $fn`; grep $z /tmp/full_files; done > /mnt/4tb_ssd/nosig_data/train_val_fullpath.txt
(spf) mouse9911@kalman:~/g

python spf/sdrpluto/sdr_controller.py --receiver-uri ip:192.168.1.17 --mode rx --fs 16000000 --fi 100000 --fc 2412000000 --rx-mode fast_attack --rx-n 524288 --rx-spacing 0.025

Segment 3p3
cat /mnt/4tb_ssd/nosig_data/train.txt | while read x; do x=$(grep $(basename $x) /tmp/full_files); python spf/scripts/segment_zarr.py --input-zarr $x --precompute-cache /mnt/4tb_ssd/precompute_cache_3p3/ --gpu -p 16; done; python spf/scripts/train_single_point.py --config latest_configs/nov28_single_config_sigma0p05_rerun_3p3.yaml ; python spf/scripts/train_single_point.py --conifg latest_configs/nov28_paired_sigma0p05_rerun_3p3.yaml

Pluto listen receive
python spf/sdrpluto/sdr_controller.py --receiver-uri ip:192.168.1.17 --mode rx --fs 16000000 --fi 100000 --fc 915000000 --rx-mode fast_attack --rx-n 524288 --rx-spacing 0.04

November 29
Segmentation metrics
~/gits/spf/spf/scripts$ python segmentation_metrics.py -i /mnt/4tb_ssd/nosig_data/*.zarr -c /mnt/4tb_ssd/precompute_cache_new --segmentation-version 3.2 > segmentation_metrics_3p2
~/gits/spf/spf/scripts$ python segmentation_metrics.py -i /mnt/4tb_ssd/nosig_data/*.zarr -c /mnt/4tb_ssd/precompute_cache --segmentation-version 3.11

python segmentation_metrics.py -i /mnt/4tb_ssd/nosig_data/train.txt -c /mnt/4tb_ssd/precompute_cache --segmentation-version 3.11 > segmentation_metrics_3p11_nov4train
python segmentation_metrics.py -i /mnt/4tb_ssd/nosig_data/train.txt -c /mnt/4tb_ssd/precompute_cache_new --segmentation-version 3.2 > segmentation_metrics_3p2_nov4train

Precompute cache
for x in /mnt/md1/2d_wallarray_v2_data/*/*.zarr /mnt/md0/spf/2d_wallarray_v2_data/oct_batch2/*.zarr; do python spf/scripts/segment_zarr.py --input-zarr $x --precompute-cache /mnt/4tb_ssd/precompute_cache_3p3/ --gpu; done

November 28
while [1 -lt 2]; do rsync -avP --remove-source-files pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.zarr' ./ ; rsync -avP pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.yaml' ./ ; for x in *.zarr; do python /home/mouse9911/gits/spf/spf/dataset/zarr_rechunk.py --no-check-contents -s $x /mnt/4tb_ssd/nosig_data/`basename $x`; python /home/mouse9911/gits/spf/spf/scripts/segment_zarr.py --input-zarr $x --precompute-cache /mnt/4tb_ssd/precompute_cache_new/ --gpu -p 1; done; sleep 600; done

November 27
Compiling for pluto
https://ez.analog.com/adieducation/university-program/f/q-a/108205/plutosdr-minimal-tool-chain-for-c-compiling

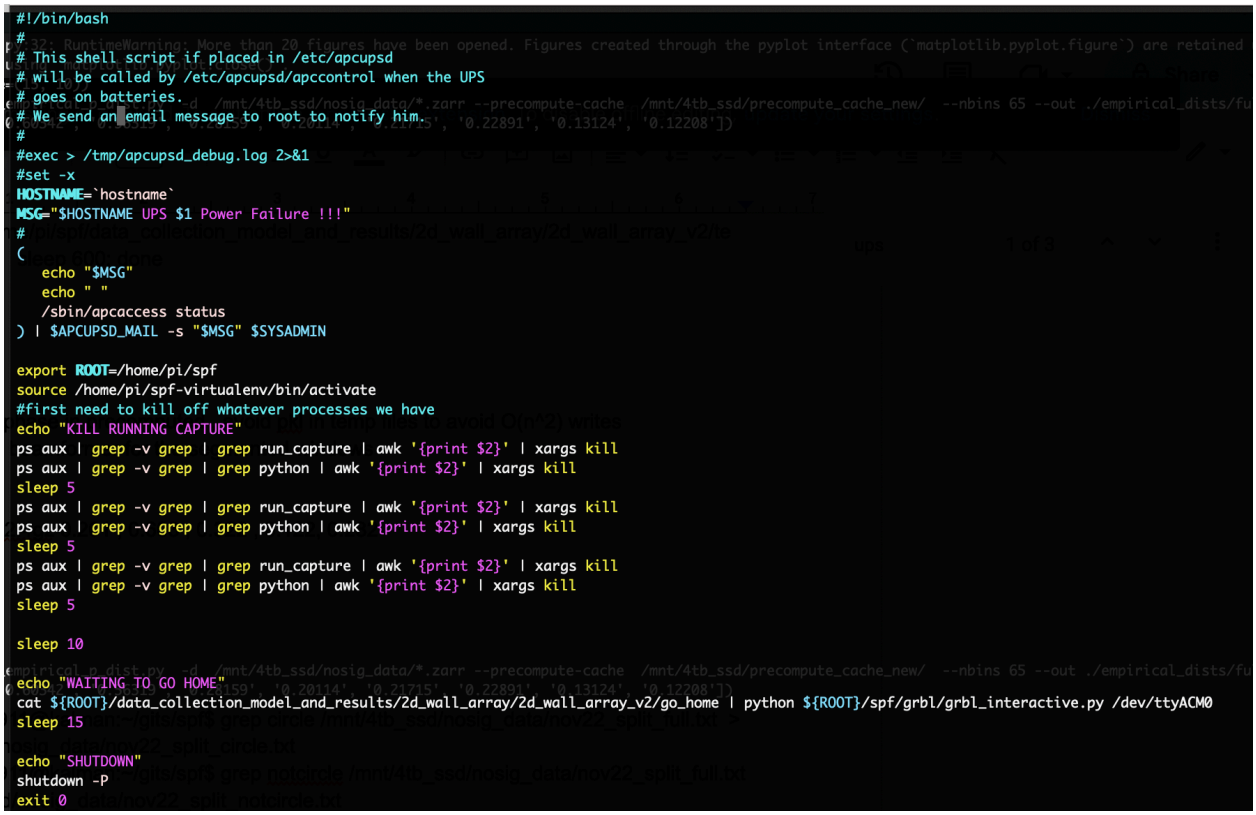
November 25
Inference cache

Create report
python spf/scripts/run_filters_report.py --output filter_reports/nov25_ --work-dir spf/run_on_filters_nov25

November 24

Overfitting on 2.4gz 0.201 , 0.603 , 0.322 , 0.122, 0.282

UPS script



```
#!/bin/bash
#
# This shell script if placed in /etc/apcupsd
# will be called by /etc/apcupsd/apccontrol when the UPS
# goes on batteries.
# We send an email message to root to notify him.
#
#exec > /tmp/apcupsd_debug.log 2>&1
#set -x
HOSTNAME=`hostname`
MSG="$HOSTNAME UPS $1 Power Failure !!!"
#
(
   echo "$MSG"
   echo " "
   /sbin/apcaccess status
) | $APCUPSD_MAIL -s "$MSG" $SYSADMIN

export ROOT=/home/pi/spf
source /home/pi/spf-virtualenv/bin/activate
#first need to kill off whatever processes we have
echo "KILL RUNNING CAPTURE"
ps aux | grep -v grep | grep run_capture | awk '{print $2}' | xargs kill
ps aux | grep -v grep | grep python | awk '{print $2}' | xargs kill
sleep 5
ps aux | grep -v grep | grep run_capture | awk '{print $2}' | xargs kill
ps aux | grep -v grep | grep python | awk '{print $2}' | xargs kill
sleep 5
ps aux | grep -v grep | grep run_capture | awk '{print $2}' | xargs kill
ps aux | grep -v grep | grep python | awk '{print $2}' | xargs kill
sleep 5

sleep 10

echo "WAITING TO GO HOME"
cat ${ROOT}/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/go_home | python ${ROOT}/spf/grbl/grbl_interactive.py /dev/ttyACM0
sleep 15

echo "SHUTDOWN"
shutdown -P
exit 0
```

Merge rover runs
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/nov20_mission1.zarr --rx ../rover1/nov20_mission1.zarr --output nov20_mission1_rover1.zarr
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/nov20_mission2.zarr --rx ../rover1/nov20_mission2.zarr --output nov20_mission2_rover1.zarr
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/nov20_mission1.zarr --rx ../rover3/nov20_mission1.zarr --output nov20_mission1_rover3.zarr
python ~/gits/spf/spf/dataset/v4_tx_rx_to_v5.py --tx ../rover2/nov20_mission2.zarr --rx ../rover3/nov20_mission2.zarr --output nov20_mission2_rover3.zarr

November 23
(spf) mouse9911@kalman:~/gits/spf$ grep circle /mnt/4tb_ssd/nosig_data/nov22_split_full.txt > /mnt/4tb_ssd/nosig_data/nov22_split_circle.txt
(spf) mouse9911@kalman:~/gits/spf$ grep notcircle /mnt/4tb_ssd/nosig_data/nov22_split_full.txt > /mnt/4tb_ssd/nosig_data/nov22_split_notcircle.txt
(spf) mouse9911@kalman:~/gits/spf$ wc -l /mnt/4tb_ssd/nosig_data/nov22_split_notcircle.txt
0 /mnt/4tb_ssd/nosig_data/nov22_split_notcircle.txt

```
(spf) mouse9911@kalman:~/gits/spf$ ^C
(spf) mouse9911@kalman:~/gits/spf$ grep -v circle /mnt/4tb_ssd/nosig_data/nov22_split_full.txt
> /mnt/4tb_ssd/nosig_data/nov22_split_notcircle.txt
(spf) mouse9911@kalman:~/gits/spf$ wc -l /mnt/4tb_ssd/nosig_data/nov22_split_notcircle.txt
737 /mnt/4tb_ssd/nosig_data/nov22_split_notcircle.txt
(spf) mouse9911@kalman:~/gits/spf$ wc -l ^C
(spf) mouse9911@kalman:~/gits/spf$ ^C
(spf) mouse9911@kalman:~/gits/spf$ head -n 663
/mnt/4tb_ssd/nosig_data/nov22_split_notcircle.txt >  /mnt/4tb_ssd/nosig_data/nov23_train.txt
(spf) mouse9911@kalman:~/gits/spf$ tail -n 74
/mnt/4tb_ssd/nosig_data/nov22_split_notcircle.txt  > /mnt/4tb_ssd/nosig_data/nov23_val.txt
(spf) mouse9911@kalman:~/gits/spf$ cat /mnt/4tb_ssd/nosig_data/nov22_split_circle.txt >>
/mnt/4tb_ssd/nosig_data/nov23_val.txt
```

November 22
RSYNC and NOSIG and PRECOMPUTE

```
while [ 1 -lt 2 ]; do  rsync -avP
pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/te
mp/*.zarr' ./ ; rsync -avP
pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/te
mp/*.yaml' ./ ; for x in *.zarr; do python /home/mouse9911/gits/spf/spf/dataset/zarr_rechunk.py
--no-check-contents -s $x /mnt/4tb_ssd/nosig_data/`basename $x`; python
/home/mouse9911/gits/spf/spf/scripts/segment_zarr.py --input-zarr $x --precompute-cache
/mnt/4tb_ssd/precompute_cache_new/ --gpu -p 1; done;  sleep 600; done
```

Create empirical dists

```
python spf/scripts/create_empirical_p_dist.py  -d  /mnt/4tb_ssd/nosig_data/*.zarr
--precompute-cache  /mnt/4tb_ssd/precompute_cache_new/  --nbins 65 --out
./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_
```

Create new splits

```
(spf) mouse9911@kalman:~/gits/spf$ head -n 750 /mnt/4tb_ssd/nosig_data/nov22_split_full.txt
> /mnt/4tb_ssd/nosig_data/nov22_split_train.txt
(spf) mouse9911@kalman:~/gits/spf$ tail -n 286 /mnt/4tb_ssd/nosig_data/nov22_split_full.txt >
/mnt/4tb_ssd/nosig_data/nov22_split_val.txt
(spf) mouse9911@kalman:~/gits/spf$ cat /mnt/4tb_ssd/nosig_data/nov22_split_val.txt
/mnt/4tb_ssd/nosig_data/nov22_split_train.txt | sort uniq | wc -l
sort: cannot read: uniq: No such file or directory
0
(spf) mouse9911@kalman:~/gits/spf$ cat /mnt/4tb_ssd/nosig_data/nov22_split_val.txt
/mnt/4tb_ssd/nosig_data/nov22_split_train.txt | sort | uniq | wc -l
1036
```

(spf) mouse9911@kalman:~/gits/spf$ cat /mnt/4tb_ssd/nosig_data/nov22_split_val.txt /mnt/4tb_ssd/nosig_data/nov22_split_train.txt | sort | wc -l
1036

November 20

Reprocess segmentation
find /mnt/md1/2d_wallarray_v2_data/ /mnt/md0/spf/2d_wallarray_v2_data/oct_batch2/ | grep "zarr$" | shuf > /tmp/toprocess
split -l 260 /tmp/toprocess /tmp/toprocess_

RSYNC
while [1 -lt 2]; do rsync -avP pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.zarr' ./ ; rsync -avP pi@devpi:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.yaml' ./ ; sleep 600; done

Post rover mission

Shrink files
for x in *zarr* ; do python ../spf/spf/dataset/zarr_shrink.py $x; done

Move to missions
 x=rover_2024_11_20_15_06_02_nRX1_circle_spacing0p05075_tag_RO2; z='nov20_mission2'; for y in ./${x}*; do ln -s $y ${z}.${y##*.}; done

November 19

Data preparation (No signal version)
for x in /mnt/md1/2d_wallarray_v2_data/oct/*.zarr; do python spf/dataset/zarr_rechunk.py -s $x /mnt/4tb_ssd/nosig_data/`basename $x`; done
for x in /mnt/md0/spf/2d_wallarray_v2_data/oct_batch2/*.zarr; do python spf/dataset/zarr_rechunk.py -s $x /mnt/4tb_ssd/nosig_data/`basename $x`; done

Data preparation (precompute cache)

for x in /mnt/md1/2d_wallarray_v2_data/oct/*.zarr; do python spf/scripts/segment_zarr.py --input-zarr $x --precompute-cache /mnt/4tb_ssd/precompute_cache/ --gpu; done
for x in /mnt/md0/spf/2d_wallarray_v2_data/oct_batch2/*.zarr; do python spf/scripts/segment_zarr.py --input-zarr $x --precompute-cache /mnt/4tb_ssd/precompute_cache/ --gpu; done

Create empirical dists
python spf/scripts/create_empirical_p_dist.py -d /mnt/4tb_ssd/nosig_data/*.zarr --precompute-cache /mnt/4tb_ssd/precompute_cache/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

Run fake drone (docker):

Run on Kalman
docker run --rm -it -p 14590-14595:14590-14595 csmisko/ardupilotspf:latest /ardupilot/Tools/autotest/sim_vehicle.py -l 37.76509485,-122.40940127,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590 --out tcpin:0.0.0.0:14591
Run ground q control - connect to IP on port 14591
Run on drone
bash ./data_collection_model_and_results/rover/rover_v3.1/drone_run.sh debug

Receiver position bearing
https://github.com/misko/spf/blob/main/spf/notebooks/wall_array_positions.ipynb

```
"""
Ceiling / Roof

    x+ <-------- O
                 |
                 |
                 |
                 |
                 V y+

R(18,0)   R(17,0)
  \         /
   \       /
    \     /
      \ /
       X
```

```
    /   \
   /     \
  /       \
R(17,1)   R(18,1)

Bearing 0 is straight towards floor

    o
    |
    |
    |
    V

Bearing 0+ is

    o
  / |
/   |
<    |
    V



Floor
"""
```

November 16

Find which files are done
find spf/run_on_filters_nov11/ | grep dual_radio_NN | sed 's@.*dual_radio_NN/\(.*.zarr\).*@\1@g' | uniq | sort | uniq

November 13
Shrink ZARR after missions
for x in *.zarr*; do python ../spf/spf/dataset/zarr_shrink.py $x; done

November 12

Rover drones and commands
ssh pi@192.168.1.43
sudo systemctl status mavlink_controller.service | tail -n 300
bash data_collection_model_and_results/rover/rover_v3.1/debug_drone_run.sh

Run full drone software
bash /home/pi/spf/data_collection_model_and_results/rover/rover_v3.1/drone_run.sh

Taranis Q controller
https://docs.google.com/presentation/d/1XHlvml56mzAEYOxwF0cEEmUB6TTZ01Ay6oxpoi7L7ws/edit?usp=sharing

Launch fake drone
docker run --rm -it -p 14590:14590 ardupilot_spf /ardupilot/Tools/autotest/sim_vehicle.py -l 37.76509485,-122.40940127,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590

Launch fake drone with multiple options to connect to it
docker run --rm -it -p 14590-14595:14590-14595 ardupilot_spf /ardupilot/Tools/autotest/sim_vehicle.py -l 37.76509485,-122.40940127,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590 --out tcpin:0.0.0.0:14591

November 11
dirs="./nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p5_from_sigma0p05 ./nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p05_outputNtheta129 ./nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p01_from_sigma0p05 ./nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p15_from_sigma0p05"; for dir in $dirs; do echo $dir; python spf/scripts/create_inference_cache.py --inference-cache /mnt/4tb_ssd/inference_cache/ --config-fn ${dir}/config.yml --checkpoint-fn ${dir}/best.pth --device cuda --datasets /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_*.zarr --parallel 24 --precompute-cache ~/precompute_cache_chunk16_sept/ ; done

cat /mnt/4tb_ssd/nosig_data/val.txt_shuf | while read x; do python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc --precompute-cache ~/precompute_cache_chunk16_sept/ --empirical-pkl-fn `pwd`/empirical_dists/full.pkl --parallel 24 --output out --work-dir spf/run_on_filters_nov11 --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml; done

November 9th
dir=./nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p05; python spf/scripts/create_inference_cache.py --inference-cache /mnt/4tb_ssd/inference_cache/ --config-fn ${dir}/config.yml --checkpoint-fn ${dir}/best.pth --device cuda --datasets /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_*.zarr --parallel 24 --precompute-cache ~/precompute_cache_chunk16_sept/

November 8th
Turn off swap
```
sudo swapoff -a
```

https://ieeexplore.ieee.org/document/10428039

November 7th
cat /mnt/4tb_ssd/nosig_data/val.txt | shuf | while read x; do python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc --precompute-cache ~/precompute_cache_chunk16_sept/ --empirical-pkl-fn `pwd`/empirical_dists/full.pkl --parallel 30 --output out --work-dir spf/run_on_filters_nov7 --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml; done

python spf/scripts/create_inference_cache.py --inference-cache /mnt/4tb_ssd/inference_cache/ --config-fn /home/mouse9911/gits/spf/nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p05/config_sigma0p1.yml --checkpoint-fn /home/mouse9911/gits/spf/nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p05/best.pth --device cuda --datasets /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_*.zarr --parallel 24 --precompute-cache ~/precompute_cache_chunk16_sept/

python spf/scripts/create_inference_cache.py --inference-cache /mnt/4tb_ssd/inference_cache/ --config-fn /home/mouse9911/gits/spf/nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p05/config_sigma0p25.yml --checkpoint-fn /home/mouse9911/gits/spf/nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p05/best.pth --device cuda --datasets /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_*.zarr --parallel 24 --precompute-cache ~/precompute_cache_chunk16_sept/

November 6th

cat /mnt/4tb_ssd/nosig_data/val.txt | shuf | while read x; do python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc --precompute-cache ~/precompute_cache_chunk16_sept/ --empirical-pkl-fn `pwd`/empirical_dists/full.pkl --parallel 30 --output out --work-dir spf/run_on_filters_nov6 --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml; done

python spf/scripts/create_inference_cache.py --inference-cache /mnt/4tb_ssd/inference_cache/ --config-fn /home/mouse9911/gits/spf/nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p05/confi

g.yml --checkpoint-fn /home/mouse9911/gits/spf/nov4_checkpoints/paired_checkpoints_inputdo0p3_sigma0p05/best.pth --device cuda --datasets /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_*.zarr --parallel 24 --precompute-cache ~/precompute_cache_chunk16_sept/

Training networks with sigma 0.05 in hopes to get better PF performance
python spf/scripts/train_single_point.py --config latest_configs/single_config_sigma0p05.yaml ; python spf/scripts/train_single_point.py --config latest_configs/paired_sigma0p05.yaml

cat /mnt/4tb_ssd/nosig_data/val.txt | shuf | while read x; do echo $x; python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc --precompute-cache ~/precompute_cache_chunk16_sept/ --empirical-pkl-fn `pwd`/empirical_dists/full.pkl --parallel 30 --output out --work-dir spf/run_on_filters_nov5 --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml; done

November 5th
Train networks
python spf/scripts/train_single_point.py --config latest_configs/single_config.yaml ; python spf/scripts/train_single_point.py --config latest_configs/paired.yaml

#run filters on val dataset
cat /mnt/4tb_ssd/nosig_data/val.txt | shuf | while read x; do python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc --precompute-cache ~/precompute_cache_chunk16_sept/ --empirical-pkl-fn `pwd`/empirical_dists/full.pkl --parallel 30 --output out --work-dir spf/run_on_filters_nov5 --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml; done

November 4th
Create new dists for wavelength numbers
python spf/scripts/create_empirical_p_dist.py -d /mnt/4tb_ssd/nosig_data/*.zarr --precompute-cache /home/mouse9911/precompute_cache_chunk16_sept/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

find `pwd`/ | grep "zarr$" | sed 's/.zarr//g' | sort | uniq | shuf > shuf_files.txt
n=`wc -l shuf_files.txt  | awk '{print $1*0.8}'`
r=`wc -l shuf_files.txt  | awk -v n=$n '{print $1-n}'`
head -n $n shuf_files.txt > train.txt
tail -n $r shuf_files.txt > val.txt

November 2nd

https://grafana.com/docs/grafana-cloud/monitor-infrastructure/integrations/integration-reference/integration-linux-node/
https://www.youtube.com/watch?v=9TJx7QTrTyo
Grafana
Monitoring

Installed grafana
Made a grafana account
On the account clicked on connections -> linux server, made a token, copy and pasted instructions to run on kalman to install alloy
Followed instructions updated alloy config on kalman
https://grafana.com/grafana/dashboards/7675-linux-servers/

November 3

Randomly go through files and run filters
```
ls  /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_*.zarr -d | shuf | while read x; do python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc  --precompute-cache ~/precompute_cache_chunk16_sept/ --empirical-pkl-fn `pwd`/empirical_dists/full.pkl --parallel 30 --output out --work-dir spf/run_on_filters_nov3 --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml ; done
```

Create report
```
python spf/scripts/run_filters_report.py --output filter_reports/report_nov3_1.csv --work-dir spf/run_on_filters_nov3
```

Run filters on data
```
for x in /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_09_*.zarr; do python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc  --precompute-cache ~/precompute_cache_chunk16_sept/ --empirical-pkl-fn `pwd`/empirical_dists/full.pkl --parallel 30 --output out --work-dir spf/run_on_filters_nov3 --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml ; done
```

Generate inference cache
```
python spf/scripts/create_inference_cache.py --inference-cache /mnt/4tb_ssd/inference_cache/ --config-fn /home/mouse9911/gits/spf/nov2_checkpoints/paired_checkpoints_inputdo0p3/config.yml --checkpoint-fn /home/mouse9911/gits/spf/nov2_checkpoints/paired_checkpoints_inputdo0p3/best.pth --device cuda --datasets /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_*.zarr  --parallel 24 --precompute-cache ~/precompute_cache_chunk16_sept/
```

November 2

for x in /mnt/4tb_ssd/nosig_data/wallarrayv3_2024_09_*.zarr; do python spf/scripts/run_filters_on_data.py -d $x --nthetas 65 --device cpu --skip-qc --precompute-cache ~/precompute_cache_chunk16_sept/ --empirical-pkl-fn empirical_dists/full.pkl --parallel 12 --output out --work-dir spf/work3/ --config ./spf/model_training_and_inference/models/ekf_and_pf_config.yml ; done

October 29
Self hosted runner
https://github.com/misko/spf/settings/actions/runners/new?arch=x64&os=linux

Download

```
# Create a folder
$ mkdir actions-runner && cd actions-runner
Copied!
# Download the latest runner package
$ curl -o actions-runner-linux-x64-2.320.0.tar.gz -L
https://github.com/actions/runner/releases/download/v2.320.0/actions-runne
r-linux-x64-2.320.0.tar.gz
Copied!
# Optional: Validate the hash
$ echo "93ac1b7ce743ee85b5d386f5c1787385ef07b3d7c728ff66ce0d3813d5f46900
actions-runner-linux-x64-2.320.0.tar.gz" | shasum -a 256 -c
Copied!
# Extract the installer
$ tar xzf ./actions-runner-linux-x64-2.320.0.tar.gz
Copied!
```

Configure

```
# Create the runner and start the configuration experience
$ ./config.sh --url https://github.com/misko/spf --token
AAH7EZY3LZ6JTRPWIMSSOW3HEGXZO
Copied!
# Last step, run it!
$ ./run.sh
```

Adding as a service

https://docs.github.com/en/actions/hosting-your-own-runners/managing-self-hosted-runners/configuring-the-self-hosted-runner-application-as-a-service

October 27
Github runners - can do a self hosted runner on kalman

October 26
Matrix sandwich covariance matrix A A transpose

2 Answers

Sorted by: Highest score (default)

2

It is a general fact that if Z is a random vector with mean μ and covariance Σ, and X is a constant matrix, then $\text{cov}(XZ) = X\Sigma X^T$. The expression you've asked about gives the Covariance for a linear transformation of a random vector.

To show this,

$$\text{cov}(XZ) = E[(XZ - X\mu)(XZ - X\mu)^T]$$

$$\text{cov}(XZ) = XE[(Z - \mu)(Z - \mu)^T]X^T = X\text{cov}(Z)X^T$$

Share Cite Follow

edited Jun 17, 2019 at 23:13

answered Jun 14, 2019 at 2:25

Brian Borchers
11.1k ● 2 ■ 25 ▲ 31

Kalman gain

$$\sigma + \sigma_z$$

I use the subscript z to denote the measurement. We can rewrite this as:

$$\mu = \left(\frac{\bar{\sigma}^2}{\bar{\sigma}^2 + \sigma_z^2}\right)\mu_z + \left(\frac{\sigma_z^2}{\bar{\sigma}^2 + \sigma_z^2}\right)\bar{\mu}$$

In this form it is easy to see that we are scaling the measurement and the prior by weights:

$$\mu = W_1\mu_z + W_2\bar{\mu}$$

The weights sum to one because the denominator is a normalization term. We introduce a new term, $K = W_1$, giving us:

$$\begin{aligned}\mu &= K\mu_z + (1 - K)\bar{\mu} \\ &= \bar{\mu} + K(\mu_z - \bar{\mu})\end{aligned}$$

where

$$K = \frac{\bar{\sigma}^2}{\bar{\sigma}^2 + \sigma_z^2}$$

K is the *Kalman gain*. It's the crux of the Kalman filter. It is a scaling term that chooses a value partway between μ_z and $\bar{\mu}$.

October24

TODO: get inference running for paired network
Run validation with and without each data type
- Only mean phase
- Only empirical
- Only beamformer

Get particle filter working
With particle filter and paired model should be able to do 2d and 3d!

October 22
FLIP multipaired data by flipping target and paired radio intermediates?

Remount in read only to get speed up
sudo mount -o remount,ro /dev/md1 /mnt/md1

Nosignal nosig rechunk
mouse9911@kalman:~/gits/spf$
for x in /mnt/md1/2d_wallarray_v2_data/oct/*.zarr; do python spf/dataset/zarr_rechunk.py -s $x /mnt/4tb_ssd/nosig_data/`basename $x`; done
Precompute cache
for x in /mnt/md1/2d_wallarray_v2_data/oct/*.zarr; do python spf/scripts/segment_zarr.py --input-zarr $x --precompute-cache /home/mouse9911/precompute_cache_chunk16_sept/ --gpu; done

Compute empirical
python spf/scripts/create_empirical_p_dist.py -d /mnt/4tb_ssd/nosig_data/*.zarr --precompute-cache /home/mouse9911/precompute_cache_chunk16_sept/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

for x in /mnt/md1/2d_wallarray_v2_data/oct/*.zarr; do python spf/scripts/segment_zarr.py $x; done

```
"/home/mouse9911/precompute_cache_chunk16_sept/"
```

CNC settings GCODE settings
$0=10
$1=25
$2=15
$3=3
$4=0
$5=0
$6=0

$10=40
$11=0.010
$12=0.002
$13=0
$20=0
$21=1
$22=1
$23=31
$24=25.000
$25=3500.000
$26=250
$27=1.000
$30=1000
$31=0
$32=0
$33=0
$34=0
$35=0
$36=0
$37=0
$38=10
$39=0
$100=2148.000
$101=2148.000
$102=2148.000
$103=2148.000
$110=175.000
$111=175.000
$112=175.000
$113=175.000
$120=100.000
$121=100.000
$122=100.000
$123=100.000
$130=1500.000
$131=1500.000
$132=1500.000
$133=1500.000
ok

October 20
https://resource.heltec.cn/download/WiFi_LoRa_32_V3/HTIT-WB32LA_V3(Rev1.1).pdf

python spf/sdrpluto/sdr_controller.py --receiver-uri ip:192.168.1.18 --mode rx --fs 16000000 --fi 100000 --fc 8680000c00 --rx-mode fast_attack --rx-n 524288

Setup LoRa

Checksum all md0 vs md1

Oct 19
RF95
https://github.com/sandeepmistry/arduino-LoRa

RF69
https://github.com/LowPowerLab/RFM69

Oct 2
Cuda on Orin

```
jetson-containers run $(autotag l4t-pytorch)
```

```
Compile protobuf
protoc --proto_path ./hudes --python_out ./hudes ./hudes/hudes.proto
```

Sept 27
Nosig rechunk
for x in /mnt/md0/spf/2d_wallarray_v2_data/sept/*.zarr; do python dataset/zarr_rechunk.py -s $x /mnt/4tb_ssd/nosig_data/`basename $x`; done

Segment
for x in sept/wallarrayv3_2024_09_*.zarr; do python ~/gits/spf/spf/scripts/segment_zarr.py $x; done

Create empirical dists

Sept 14
python ~/gits/spf/spf/scripts/create_empirical_p_dist.py -d /mnt/md0/spf/2d_wallarray_v2_data/june_fix_nosig/*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --nbins 65 --out ./empirical_dists/full.pkl --output-fig-prefix=./empirical_dists/empirical_dists_

Sept 11

```
https://catalog.ngc.nvidia.com/orgs/nvidia/containers/l4t-pytorch
sudo docker pull nvcr.io/nvidia/l4t-pytorch:r35.2.1-pth2.0-py3
```

```
sudo docker run -it --rm --runtime nvidia --network host
nvcr.io/nvidia/l4t-pytorch:r35.2.1-pth2.0-py3

mouse9911@orin:~/gits/jetson-containers$ jetson-containers run
$(autotag l4t-pytorch)
Namespace(packages=['l4t-pytorch'], prefer=['local', 'registry',
'build'], disable=[''], user='dustynv', output='/tmp/autotag',
quiet=False, verbose=False)
-- L4T_VERSION=36.3.0  JETPACK_VERSION=6.0  CUDA_VERSION=12.2
-- Finding compatible container image for ['l4t-pytorch']
[sudo] password for mouse9911:
dustynv/l4t-pytorch:r36.2.0
+ sudo docker run --runtime nvidia -it --rm --network host --volume
/tmp/argus_socket:/tmp/argus_socket --volume
/etc/enctune.conf:/etc/enctune.conf --volume
/etc/nv_tegra_release:/etc/nv_tegra_release --volume
/tmp/nv_jetson_model:/tmp/nv_jetson_model --volume
/var/run/dbus:/var/run/dbus --volume
/var/run/avahi-daemon/socket:/var/run/avahi-daemon/socket --volume
/var/run/docker.sock:/var/run/docker.sock --volume
/home/mouse9911/gits/jetson-containers/data:/data --device /dev/snd
--device /dev/bus/usb --device /dev/i2c-0 --device /dev/i2c-1
--device /dev/i2c-2 --device /dev/i2c-4 --device /dev/i2c-5 --device
/dev/i2c-7 --device /dev/i2c-9 -v /run/jtop.sock:/run/jtop.sock
dustynv/l4t-pytorch:r36.2.0
root@orin:/# python3
Python 3.10.12 (main, Nov 20 2023, 15:14:05) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more
information.
>>> import torch
>>> torch.distributed
<module 'torch.distributed' from
'/usr/local/lib/python3.10/dist-packages/torch/distributed/__init__.p
y'>
>>> torch.distributed.is_available
<function is_available at 0xffff1ddade10>
>>> torch.distributed.is_available()
True
>>>

jetson-containers run --mount
type=bind,source=/home/mouse9911/gits/,target=/gits $(autotag
l4t-pytorch)
```

Aug 16
Train beamnet base model
https://wandb.ai/projectspf/projectspf_aug13/runs/c3jq90o9?nw=nwusercsmisko

python simple_train_filter.py -d /mnt/md0/spf/2d_wallarray_v2_data/june_fix/*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 0 --batch 256 --device cuda --wandb-project projectspf_aug13 --lr 0.0000001 --val-holdout-fraction 0.3 --val-subsample-fraction 1.0 --tformer-dmodel 128 --tformer-dhid 128 --snapshots-per-session 1 --head-start 200000 --amp --snapshots-stride 1 --tformer-dropout 0.0 --tformer-snapshot-dropout 0.0 --tformer-layers 2 --only-beamnet

Copy to nvme
https://github.com/jetsonhacks/rootOnNVMe
Flash to nvme (after above script add this to /etc/exlinux/*.conf)
APPEND ${cbootargs} quiet root=/dev/nvme0n1p1 rw rootwait rootfstype=ext4 console=ttyTCU0,115200n8 console=tty0 fbcon=map:0 net.ifnames=0

Aug 14
Jetson container
https://github.com/dusty-nv/jetson-inference/blob/master/docs/aux-docker.md

Copy to nvme
https://github.com/jetsonhacks/rootOnNVMe

Docker jetson
https://developer.nvidia.com/embedded/learn/tutorials/jetson-container

python simple_train_filter.py -d /mnt/md0/spf/2d_wallarray_v2_data/june_fix/wallarrayv3_2024_06*e.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 0 --batch 12 --device cuda --wandb-project projectspf_aug13 --lr 0.000001 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015 --tformer-dmodel 1024 --tformer-dhid 256 --snapshots-per-session 1024 --head-start 20000 --amp

Aug 13
Train base model
python simple_train_filter.py -d /mnt/md0/spf/2d_wallarray_v2_data/june_fix/wallarrayv3_2024_06*e.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 8 --batch 12 --device cuda --wandb-project projectspf_jul9 --lr 0.000001 --val-holdout-fraction 0.3 --val-subsample-fraction

0.015 --tformer-dmodel 1024 --tformer-dhid 256 --snapshots-per-session 1024 --head-start 20000 --amp

for x in *.zarr; do python ~/gits/spf/spf/notebooks/segment_zarr.py $x; done
https://developer.nvidia.com/embedded/jetson-linux
^ jetson developer guide jetson image sdkmanager
https://www.youtube.com/watch?v=q4fGac-nrTI&t=378s
https://forums.developer.nvidia.com/t/l4t-apply-binaries-sh-error/111153/7

https://forum.seeedstudio.com/t/a603-orin-nano-8gb-jetpack-6-0-flash-issues/279632

August 8
https://github.com/dusty-nv/jetson-containers/blob/master/packages/pytorch/build.sh

August 7
rm -rf ./wallarrayv3_2024_06_03_00_30_00_nRX2_rx_circle*; time python ~/gits/spf/spf/dataset/zarr_rechunk.py ../june/wallarrayv3_2024_06_03_00_30_00_nRX2_rx_circle.zarr ./wallarrayv3_2024_06_03_00_30_00_nRX2_rx_circle.zarr; cp ../june/wallarrayv3_2024_06_03_00_30_00_nRX2_rx_circle.yaml ./; time python ~/gits/spf/spf/dataset/benchmark_dataset.py ./wallarrayv3_2024_06_03_00_30_00_nRX2_rx_circle.zarr

python simple_train_filter.py -d /mnt/4tb_ssd/june_fix/wallarrayv3_2024_06*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 12 --batch 8 --device cuda --wandb-project projectspf_jul27 --lr 0.0000005 --lr-beamnet 0.0 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015 --tformer-dmodel 1024 --tformer-dhid 256 --snapshots-per-session 1024 --head-start 0 --dtype float32 --amp --beamformer-input --no-include-input --load-beamnet this_model__step130000.chkpnt

August 5
Patch pytorch https://github.com/open-mmlab/mmdeploy/issues/2586

Aug 3
https://forums.developer.nvidia.com/t/pytorch-for-jetson/72048

sudo pip3 install -U jetson-stats
sudo apt install nvidia-jetpack

https://pytorch.org/audio/stable/build.jetson.html

https://forums.developer.nvidia.com/t/cuda-runtime-on-jetson-orin-agx/260735

python3 -m pip install --upgrade pip; python3 -m pip install numpy=='1.26.1';

~~wget~~

~~https://developer.download.nvidia.cn/compute/redist/jp/v60/pytorch/torch-2.4.0a0+3bcc3cddb5.nv24.07.16234504-cp310-cp310-linux_aarch64.whl~~

~~python3 -m pip install --no-cache~~

~~torch-2.4.0a0+3bcc3cddb5.nv24.07.16234504-cp310-cp310-linux_aarch64.whl~~

wget

https://developer.download.nvidia.com/compute/redist/jp/v60/pytorch/torch-2.4.0a0+07cecf4168.nv24.05.14710581-cp310-cp310-linux_aarch64.whl

```
python3 -m pip install --no-cache
```

torch-2.4.0a0+07cecf4168.nv24.05.14710581-cp310-cp310-linux_aarch64.whl

python3 -m venv --system-site-packages ../spf_venv

https://forums.developer.nvidia.com/t/pytorch-for-jetson/72048

Build from source

https://forums.developer.nvidia.com/t/pytorch-installation-error/275517

```
git clone --recursive --branch v2.4 http://github.com/pytorch/pytorch
sudo apt-get install python3-pip cmake libopenblas-dev libopenmpi-dev -y
cd pytorch
pip3 install -r requirements.txt
pip3 install scikit-build
pip3 install ninja
sudo nvpmodel -m 0
sudo jetson_clocks
export USE_NCCL=0
export USE_DISTRIBUTED=1
export USE_QNNPACK=0
export USE_PYTORCH_QNNPACK=0
export TORCH_CUDA_ARCH_LIST=8.7
export PYTORCH_BUILD_VERSION=2.4
export PYTORCH_BUILD_NUMBER=1
export PATH=/usr/local/cuda/bin${PATH:+:${PATH}}
export LD_LIBRARY_PATH=/usr/local/cuda/lib64\
```

```
                            ${LD_LIBRARY_PATH:+:${LD_LIBRARY_PATH}}
MAX_JOBS=5 python3 setup.py bdist_wheel
#MAX_JOBS=n python setup.py build
```

Aug1
Use Qengineering image with pytorch
python3.8 -m venv --system-site-packages ../spf_venv
Python -m pytest tests

Juy 31
python simple_train_filter.py -d /mnt/4tb_ssd/june_fix/wallarrayv3_2024_06*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 12 --batch 8 --device cuda --wandb-project projectspf_jul27 --lr 0.000001 --lr-beamnet 0.0001 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015 --tformer-dmodel 1024 --tformer-dhid 256 --snapshots-per-session 1024 --head-start 200000 --dtype float32 --amp --beamformer-input

Jetson nano pytorch install
https://developer.download.nvidia.com/assets/embedded/secure/jetson/Nano/docs/NV_Jetson_Nano_Developer_Kit_User_Guide.pdf?pExB1BUQLdaIA1FoJx1_vSoVfo09CtvLRVZptkr8yihF4uHbYKUtoYe5rjlAHCaZVGWe2gq7ak2n8BZEWNCJ93MdJo6YRYBzacNPqNHqWrvyVMGQkfkEnh4UoUkQ6NMlWYj16-h8_mIPZ58R9LKqyM6XoPcr4c2M4uuaY0bV54yV-mSfIldpCJBp0O6Ji43XD4w=&t=eyJscyI6ImdzZW8iLCJsc2QiOiJodHRwczovL3d3dy5nb29nbGUuY29tLyJ9

https://forums.developer.nvidia.com/t/pytorch-for-jetson/72048
https://elinux.org/Jetson_Zoo (downoad wheel from here?)
`sudo apt-get install python3-pip`
https://forums.developer.nvidia.com/t/jetson-nano-developer-kit/161188/2
https://forums.developer.nvidia.com/t/issues-installing-pytorch-through-jetson-inference-install-script/268123/2
`pip3 install --upgrade pip`

Jetson has older version of CUDA can only support up to pytorch 1.10!
https://forums.developer.nvidia.com/t/pytorch-for-jetson/72048
https://docs.nvidia.com/deeplearning/frameworks/install-pytorch-jetson-platform/index.html

Install jetson nano python
https://jetsonhacks.com/2023/06/12/upgrade-python-on-jetson-nano-tutorial/

https://github.com/conda-forge/miniforge
conda create --name "spf" python=3.11

Jul 30

python simple_train_filter.py -d /mnt/4tb_ssd/june_fix/wallarrayv3_2024_06*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 12 --batch 8 --device cuda --wandb-project projectspf_jul27 --lr 0.000001 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015 --tformer-dmodel 1024 --tformer-dhid 256 --snapshots-per-session 1024 --head-start 200000 --dtype float32 --amp --beamformer-input --beam-net-lambda 10000

July 28
Particle filter from temporary file
Test case for this

July 27
python simple_train_filter.py -d /mnt/4tb_ssd/june_fix/wallarrayv3_2024_06*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 8 --batch 8 --device cuda --wandb-project projectspf_jul27 --lr 0.000001 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015 --tformer-dmodel 1024 --tformer-dhid 256 --snapshots-per-session 1024 --head-start 20000 --beamformer-input --dtype float32

Long run
python simple_train_filter.py -d /mnt/4tb_ssd/june_fix/wallarrayv3_2024_06*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 8 --batch 12 --device cuda --wandb-project projectspf_jul9 --lr 0.000001 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015 --tformer-dmodel 1024 --tformer-dhid 256 --snapshots-per-session 1024 --head-start 20000 --amp

========== short test summary info ==========
FAILED tests/test_models.py::test_beamnet_downsampled - RuntimeError: expand(torch.IntTensor{[1, 1]}, size=[1]): the number of sizes provided (1) must be greater or equal to the number of dimensions in the tensor (2)
FAILED tests/test_models.py::test_beamnet_direct_positional - RuntimeError: expand(torch.IntTensor{[1, 1]}, size=[1]): the number of sizes provided (1) must be greater or equal to the number of dimensions in the tensor (2)

FAILED tests/test_models.py::test_beamnet_direct_positional_and_symmetry - RuntimeError: expand(torch.IntTensor{[1, 1]}, size=[1]): the number of sizes provided (1) must be greater or equal to the number of dimensions in the tensor (2)
FAILED tests/test_models.py::test_beamnet_direct - RuntimeError: expand(torch.IntTensor{[1, 1]}, size=[1]): the number of sizes provided (1) must be greater or equal to the number of dimensions in the tensor (2)
FAILED tests/test_models.py::test_beamnet_discrete - RuntimeError: expand(torch.IntTensor{[1, 1]}, size=[1]): the number of sizes provided (1) must be greater or equal to the number of dimensions in the tensor (2)
FAILED tests/test_models.py::test_beamnet_discrete_symmetry - RuntimeError: expand(torch.IntTensor{[1, 1]}, size=[1]): the number of sizes provided (1) must be greater or equal to the number of dimensions in the tensor (2)
FAILED tests/test_models.py::test_beamnet_discrete_symmetry_and_positional - RuntimeError: expand(torch.IntTensor{[1, 1]}, size=[1]): the number of sizes provided (1) must be greater or equal to the number of dimensions in the tensor (2)
========== 7 failed, 37 passed,

Test case for temp dataset
Test case for particle filter

July 17

- Simulator and non linear flight paths
- Demo
- Bump torch version to fix issue in tests

July 16
Real time PF(?)
MSE is computed differently

- For single radio its single point
- Its normed by full pi in PF but half pi in NN

python grbl_radio_collection.py -c v5_configs/wall_array_v2_external_fake.yaml -r bounce
Generate fake dataset

July 14
python simple_train_filter.py -d /mnt/4tb_ssd/june_fix/wallarrayv3_2024_06*14*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 12 --batch 8 --device cuda --wandb-project projectspf_jul14 --lr 0.000001 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015 --tformer-dmodel 512 --tformer-dhid 128 --snapshots-per-session 1024 --beamnet-latent 4

July 10

`Persistent_workers`

July:
python simple_train_filter.py -d /mnt/4tb_ssd/june_fix/wallarrayv3_2024_06*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 8 --batch 8 --device cuda --lr 0.000001 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015

July 8
https://github.com/TheJacksonLaboratory/zarrdataset/tree/main/zarrdataset
Transformer token dropout
https://discuss.pytorch.org/t/iterable-dataset-with-zarr-as-storage-backend-results-in-degrading-performance/91100
Reopen lmdb!
Pin memory function in collater

July 7
python simple_train_filter.py -d /mnt/md0/spf/2d_wallarray_v2_data/june_fix/wallarrayv3_2024_06*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 8 --batch 8 --device cuda --wandb-project projectspf --lr 0.000001 --val-holdout-fraction 0.3 --val-subsample-fraction 0.015

July 3
python simple_train_filter.py -d /mnt/md0/spf/2d_wallarray_v2_data/june_fix/wallarrayv3_2024_06*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 8 --batch 8 --device cuda --wandb-project projectspf --lr 0.000001 --val-holdout-fraction 0.001

https://wandb.ai/projectspf/projectspf/runs/i8bgswwm?nw=nwusercsmisko

July 1
Pytorch transformer mask
https://stackoverflow.com/questions/62170439/difference-between-src-mask-and-src-key-padding-mask/68396781#68396781

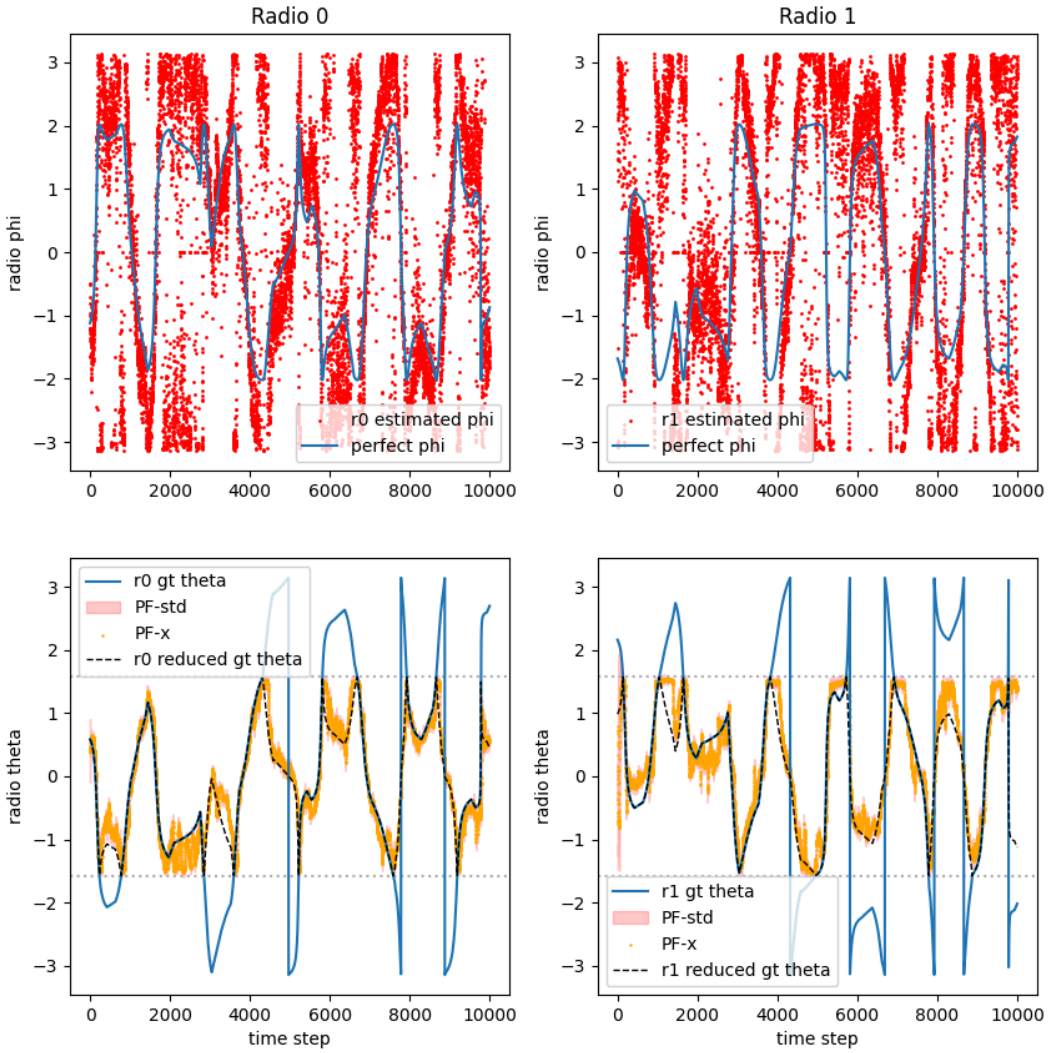
June 30:
python simple_train_filter.py -d /mnt/md0/spf/2d_wallarray_v2_data/june_fix/wallarrayv3_2024_06_*.zarr --precompute-cache ~/precompute_cache_chunk16_fresh/ --act leaky --skip-qc --workers 10 --batch 16 --device cuda --wandb-project projectspf --lr 0.000001 --val-holdout-fraction 0.01

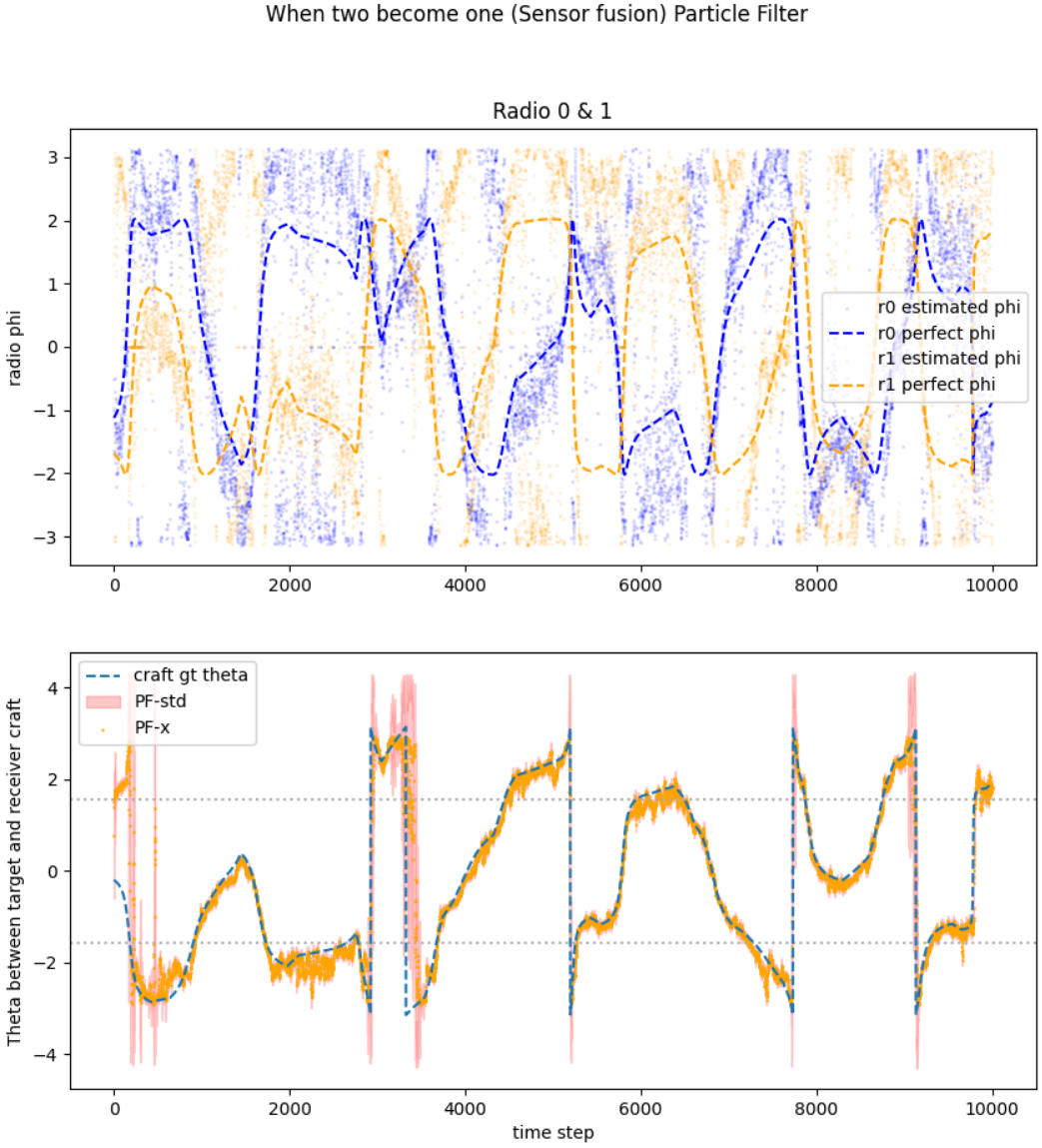
June 29:

https://github.com/Yusufma03/pfrnns?tab=readme-ov-file
https://github.com/stanford-iprl-lab/torchfilter

June 26:

Single ladies (radios) ParticleFilter





June 25:

Todo:

V4 dataset

Particle filter using distribution

June 24:

-rw-r--r--@ 1 miskodzamba staff 8788784 Jun 5 10:26 rx_antenna_holder - antenna_mount_smoothed (65mm).stl

-rw-r--r--@ 1 miskodzamba staff 9680484 Jun 5 10:27 rx_antenna_holder - antenna_mount_smoothed (40mm).stl
-rw-r--r--@ 1 miskodzamba staff 9670084 Jun 5 10:27 rx_antenna_holder - antenna_mount_smoothed (50.75mm).stl

Antenna spacing

June 20:

https://pypfilt.readthedocs.io/en/latest/

https://github.com/hilearn/pypfilt-RandomWalk/blob/master/Pypfilt-RadnomWalk-Example.ipynb

June 19:

COwork with Jason

EKF3 from GPS can we get error estimate

Tmux

Gist command line

When tracking in full theta space , need to collapse to half pi space

To not pick one of the two peaks can use beamformer as input

Z_k is 65 dim vector

Plot real data with different nomralization techniques to figure out how to write H, h

Different possible time steps if different time between observations

Add test case for `reduce_theta_to_positive_y`

```
TODOTODO f(x) needs to flip sign on velocity when it hits boundary
Restrict x to -pi/2 , pi/2 for single radio model
Symmetry in solution space
[x , dot x ] , [sign(x) , - dot x]
```

```
dt = 1.0  # 0.1
rk.F = np.eye(2) + np.array([[0, 1], [0, 0]]) * dt

phi_std = 0.5
rk.R = phi_std**2  # np.diag([phi_std**2])
# TODO R:
# Use gt state x to get a good R, then worry about how to use estimated x
# Motion model , f , wrap around
# make class for motion model and measurement model
# Plot how x and p evolve
rk.Q = Q_discrete_white_noise(2, dt=dt, var=0.3)
rk.P *= 5  # initialized as identity?
```

```
Use beam former to confirm peaks are the same
Use beam former widths to try and estimate R
```

https://rsync.net/pricing.html

Borg backup

Is is possible to have h(x) return two values and choose the one thats closer to z to minimize y?

June 18

Tailscale for vpn access reverse ssh
https://www.jessesquires.com/blog/2023/12/16/macbook-notch-and-menu-bar-fixes/

June 16:
mouse9911@kalman:~$ sudo pwrstat -status

The UPS information shows as following:

Properties:
Model Name..... CP1500PFCLCDa
Firmware Number..... CR01802F5C11
Rating Voltage..... 120 V
Rating Power..... 1000 Watt(1500 VA)

Current UPS status:
State..... Normal
Power Supply by..... Utility Power
Utility Voltage..... 120 V
Output Voltage..... 120 V
Battery Capacity..... 100 %
Remaining Runtime..... 9 min.
Load..... 690 Watt(69 %)
Line Interaction..... None
Test Result..... Unknown
Last Power Event..... None

python spf/notebooks/simple_train.py --device cuda --seed 0 --nthetas 65 --datasets /mnt/md0/spf/2d_wallarray_v2_data/june/*circle*.zarr --batch 1024 --workers 12 --act leaky --shuffle --segmentation-level downsampled --type direct --seg-net conv --epochs 1200 --shuffle --no-sigmoid --block --n-radios 2 --skip-qc --wandb-project test_full2 --val-every 500 --val-holdout-fraction 0.15 --lr 0.0001 --paired-start 0 --seg-start 0 --paired-lambda 0.2 --depth 8 --hidden 256 --batch-norm --precompute-cache ~/precompute_cache_chunk16_fresh/ --other (running but looks ok)

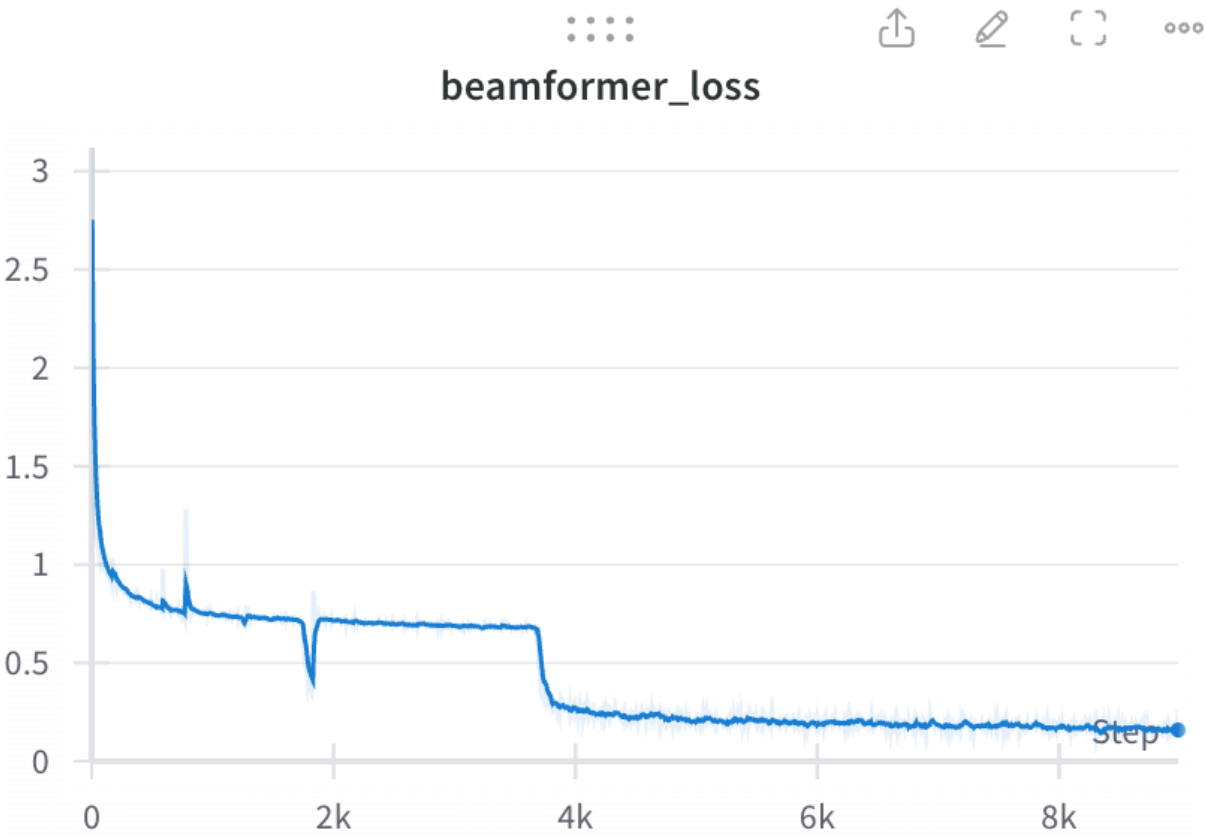
Antenna spacing is very important up to sign! This can flip phi estimates for right/left of array! Rx_spacing negative -np.sin

6ee5e28f78f1ce132ba6a9e6552c7a6a7226dd95
python spf/notebooks/simple_train.py --device cuda --seed 0 --nthetas 65 --datasets /mnt/md0/spf/2d_wallarray_v2_data/june/*circle*.zarr --batch 1024 --workers 12 --act leaky --shuffle --segmentation-level downsampled --type direct --seg-net conv --epochs 1200 --shuffle --no-sigmoid --block --n-radios 1 --skip-qc --wandb-project test_full --val-every 500 --val-holdout-fraction 0.15 --lr 0.0001 --paired-start 0 --seg-start 0 --paired-lambda 0.2 --depth 8 --hidden 256 --batch-norm --precompute-cache ~/precompute_cache_chunk16_fresh/ --other

https://stackoverflow.com/questions/21245167/kalman-filter-in-computer-vision-the-choice-of-q-and-r-noise-covariances
Q R noise matrix ekf kalman

Baseline seems to work
python spf/notebooks/simple_train.py --device cuda --seed 0 --nthetas 65 --datasets /mnt/md0/spf/2d_wallarray_v2_data/june/*.zarr --batch 1024 --workers 12 --act leaky --shuffle --segmentation-level downsampled --type direct --seg-net conv --epochs 1200 --shuffle --symmetry --no-sigmoid --block --n-radios 2 --skip-qc --wandb-project test123 --val-every 500 --val-holdout-fraction 0.15 --lr 0.001 --paired-start 0 --seg-start 0 --paired-lambda 0.2 --depth 16 --hidden 256 --batch-norm --precompute-cache ~/precompute_cache_chunk16_fresh/

–other really kicks baselines ass
python spf/notebooks/simple_train.py --device cuda --seed 0 --nthetas 65 --datasets /mnt/md0/spf/2d_wallarray_v2_data/june/*circle*.zarr --batch 1024 --workers 12 --act leaky --shuffle --segmentation-level downsampled --type direct --seg-net conv --epochs 1200 --shuffle --no-sigmoid --block --n-radios 1 --skip-qc --wandb-project test_full --val-every 500 --val-holdout-fraction 0.15 --lr 0.001 --paired-start 0 --seg-start 0 --paired-lambda 0.2 --depth 8 --hidden 256 --batch-norm --precompute-cache ~/precompute_cache_chunk16_fresh/ --other



June 15
bash data_collection_model_and_results/rover/rover_v3.1/debug_drone_run.sh

June 13
These are called *state-space* methods because we are expressing the solution of the differential equations in terms of the system state.

State-Space Representation of Dynamic Systems

We've derived the equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{w}$$

However, we are not interested in the derivative of $\mathbf{x}$, but in $\mathbf{x}$ itself. Ignoring the noise for a moment, we want an equation that recursively finds the value of $\mathbf{x}$ at time t_k in terms of $\mathbf{x}$ at time t_{k-1}:

$$\mathbf{x}(t_k) = \mathbf{F}(\Delta t)\mathbf{x}(t_{k-1}) + \mathbf{B}(t_k)\mathbf{u}(t_k)$$

Convention allows us to write $\mathbf{x}(t_k)$ as $\mathbf{x}_k$, which means the the value of $\mathbf{x}$ at the k^{th} value of t.

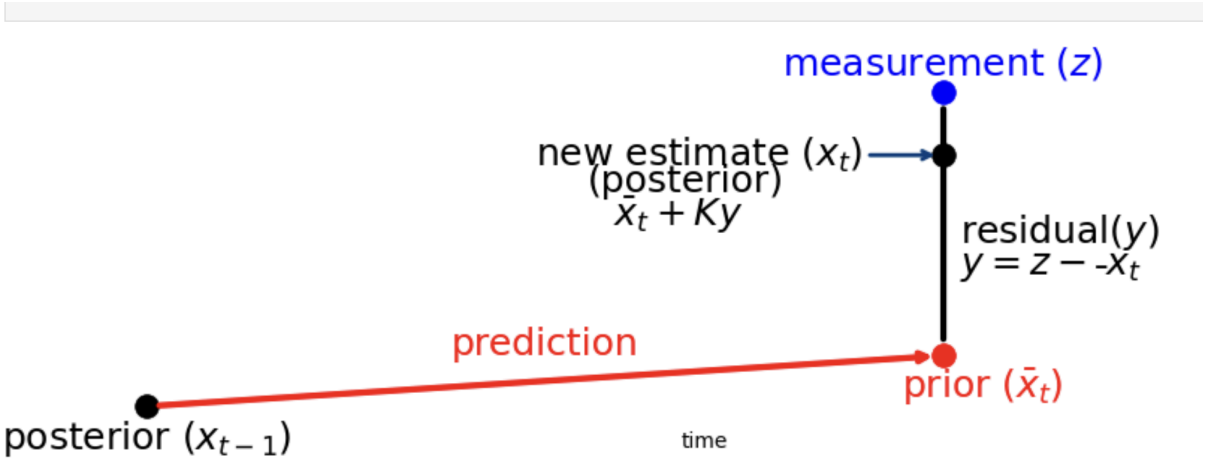
$$\mathbf{x}_k = \mathbf{F}\mathbf{x}_{k-1} + \mathbf{B}_k\mathbf{u}_k$$

$\mathbf{F}$ is the familiar *state transition matrix*, named due to its ability to transition the state's value between discrete time steps. It is very similar to the system dynamics matrix $\mathbf{A}$. The difference is that $\mathbf{A}$ models a set of linear differential equations, and is continuous. $\mathbf{F}$ is discrete, and represents a set of linear equations (not differential equations) which transitions $\mathbf{x}_{k-1}$ to $\mathbf{x}_k$ over a discrete time step Δt.

```
def run(ys):
    burnin = 50
    kf = KalmanFilter(dim_x=1, dim_z=1)
    kf.F = np.array([[1.]])
    kf.H = np.array([[1.]])

    # "Fit" the filter ... library does not seem to provide this functional
    q_var = np.var(ys[:burnin])
    kf.Q = np.array([[q_var]])
    kf.R = np.array([[q_var]])

    x_rest = list()
    kf.x = np.array([0])
    kf.P *= 10.
    ys_test = ys[burnin:]
    for y in ys_test[:-1]:
        kf.predict()            # Predict does not return a value. Grumble
        x_rest.append(kf.x[0])
        kf.update([y])

    return [0] + list(ys[:burnin]) + x_rest

XS = run(YS)
len(XS)
```

June 12
https://github.com/Nuand/bladeRF/wiki

June 11
sudo gpasswd -a $USER docker newgrp docker
Fix docker error



The Kalman gain K is a scale factor that chooses a value along the residual. This leads to an alternative but equivalent i

June 8

285ed35ece4b88cdaddf825cd589190b44d66559

(spf) miskodzamba@Miskos-MBP spf % python spf/notebooks/simple_train.py --device mps --seed 0 --nthetas 11 --datasets /Volumes/SPFData2/missions/april5/wallarrayv3_2024_06_0[3456]*.zarr --batch 512 --workers 0 --act leaky --shuffle --segmentation-level downsampled --type direct --seg-net conv --epochs 1200 --shuffle --symmetry --no-sigmoid --block --n-radios 2 --skip-qc --wandb-project test123 --val-every 500 --val-holdout-fraction 0.05 --lr 0.001

June 7:

python spf/notebooks/simple_train.py --device mps --seed 0 --nthetas 11 --datasets /Volumes/SPFData2/missions/april5/wallarrayv3_2024_06_0[3456]*.zarr --batch 512 --workers 0 --act leaky --shuffle --segmentation-level downsampled --type direct --seg-net conv --epochs 1200 --shuffle --symmetry --no-sigmoid --block --n-radios 2 --skip-qc --wandb-project test123 --val-every 500 --val-holdout-fraction 0.05 --lr 0.001

June 6:

https://docs.google.com/presentation/d/1EikEzaetmG5D1mbj27Riba2iez71l92aqH70HTDbWbE/edit?usp=sharing

Jason meet up (slides)

Jason:

Ipython embed() , from ipython import embed

Nearly constant velocity or acceleration KALMAN filter for taking input of beamformer and tracking/outputting theta

Detector can rotate but subtract it out

Beam forming at 10hz , coming off the device directly (FPGA)

How fast is sampling, can we use smaller windows

Can i get multiple buffers at once

Task2: LSTM

Stationary target , moving detector, kalman filter (taking input of the theta-filter and tracking/outputting x-y position)

Extended kalman filter can have non linear function but does not have guarantee to be best

June 5

```
python spf/notebooks/simple_train.py --device mps --seed 0 --nthetas 11 --datasets /Volumes/SPFData2/missions/april5/wallarrayv3_2024_06_0[34]*.zarr --batch 128 --workers 0 --act leaky --shuffle --segmentation-level downsampled --type direct --seg-net conv --epochs 1200 --shuffle --symmetry --no-sigmoid --block --n-radios 2 --skip-qc --wandb-project test123 --val-every 500 --val-holdout-fraction 0.05

for x in /Volumes/SPFData2/missions/april5/*.zarr; do pkl="${x%.*}_segmentation.pkl"; if [ ! -f $pkl ]; then python spf/notebooks/segment_zarr.py $x; fi ; done

while [ 1 -lt 2 ]; do  ; rsync -avP pi@devpi.local:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.zarr' ./ ; rsync -avP pi@devpi.local:'/home/pi/spf/data_collection_model_and_results/2d_wall_array/2d_wall_array_v2/temp/*.yaml' ./ ; rsync -avP /Volumes/SPFData/missions/april5/* /Volumes/SPFData2/missions/april5; for x in *.zarr; do python ~/Dropbox/research/gits/spf/spf/notebooks/segment_zarr.py $x; done; sleep 600; done
```

June 3

Old grbl config:

```
$0=10
$1=25
$2=15
$3=3
$4=0
$5=0
$6=0
$10=40
$11=0.010
$12=0.002
$13=0
$20=0
$21=1
$22=1
$23=31
$24=25.000
```

$25=3500.000
$26=250
$27=1.000
$30=1000
$31=0
$32=0
$33=0
$34=0
$35=0
$36=0
$37=0
$38=10
$39=0
$100=2148.000
$101=2148.000
$102=2148.000
$103=2148.000
$110=350.000
$111=350.000
$112=350.000
$113=350.000
$120=100.000
$121=100.000
$122=100.000
$123=100.000
$130=1500.000
$131=1500.000
$132=1500.000
$133=1500.000

$110=175.000
$111=175.000
$112=175.000
$113=175.000

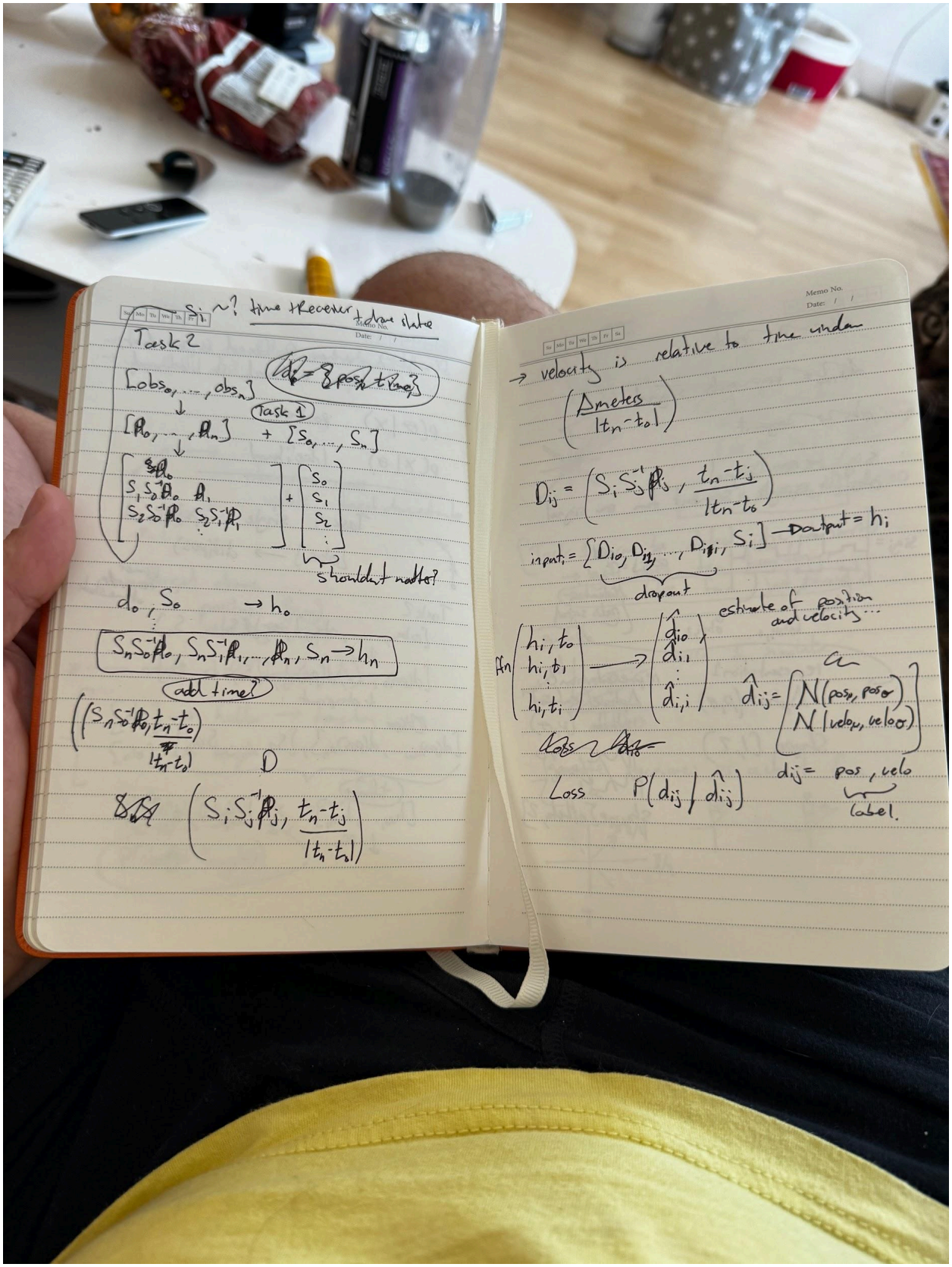
June 2
https://github.com/analogdevicesinc/plutosdr_scripts/tree/master

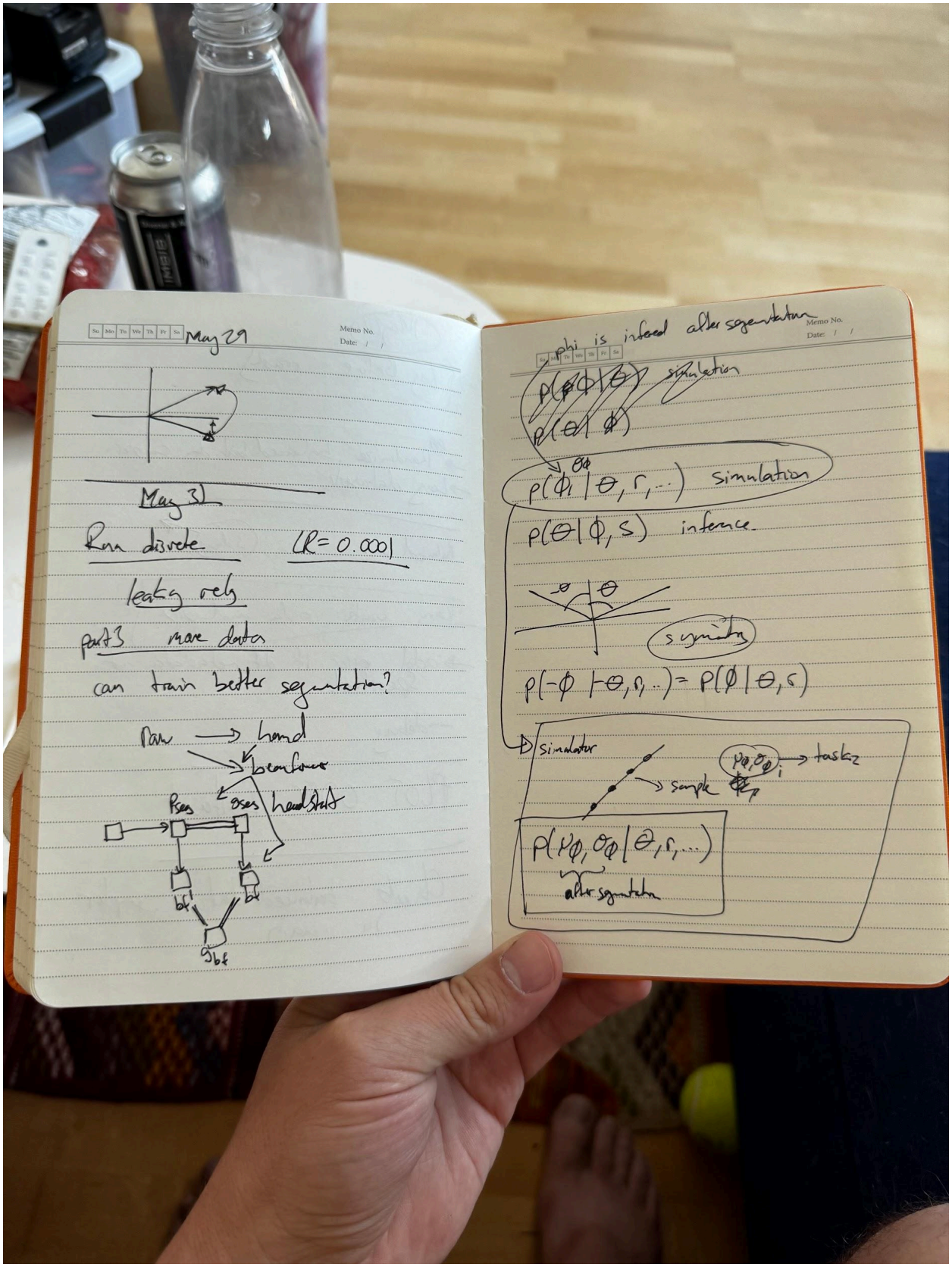
June 1
Debug
python spf/sdrpluto/sdr_controller.py --receiver-uri ip:192.168.1.17 --mode rx --fs 16000000 --fi 100000 --fc 2412000000 --rx-mode fast_attack --rx-n 100000

192.168.1.18 -> Rev C

https://www.quantulum.co.uk/blog/private-lte-with-plutoplus-sdr/ pluto plus 0.38 firmware?
https://github.com/pgreenland/plutosdr-fw/releases/tag/v0.38_plutoplus_with_timestamping

May 31
ANtenna shielding
https://www.youtube.com/watch?v=sAjWJbZOq6I&t=938s gps blocking jamming spoofing
Transformer task task1 task2


S_i ~? time + Receiver + drone state
Task 2
$[obs_0, \ldots, obs_n]$
Task 1
$[R_0, \ldots, R_n]$ + $[S_0, \ldots, S_n]$
should not matter?
$d_0, S_0 \rightarrow h_0$
$S_n S_0^{-1} R_0, S_n S_1^{-1} R_1, \ldots, R_n, S_n \rightarrow h_n$
add time?
$\left(S_i S_j^{-1} R_j, \frac{t_n - t_j}{|t_n - t_0|}\right)$
$D_{ij} = \left(S_i S_j^{-1} R_j, \frac{t_n - t_j}{|t_n - t_0|}\right)$
velocity is relative to time under
$\left(\frac{\Delta \text{meters}}{|t_n - t_0|}\right)$
input$_i$ = $[D_{i0}, D_{ij}, \ldots, D_{ii}, S_i]$ → output = h_i
dropout
$fn\begin{pmatrix} h_i, t_0 \\ h_i, t_1 \\ h_i, t_i \end{pmatrix} \rightarrow \begin{pmatrix} \hat{d}_{i0} \\ \hat{d}_{i1} \\ \hat{d}_{ii} \end{pmatrix}$
estimate of position and velocity...
$\hat{d}_{ij} = \begin{bmatrix} N(pos_\mu, pos_\sigma) \\ N(velo_\mu, velo_\sigma) \end{bmatrix}$
Loss $P(d_{ij} \mid \hat{d}_{ij})$
d_{ij} = pos, velo
label.


May 29
May 31
Run discrete
LR = 0.0001
leaky relu
part 3 more data
can train better segmentation?
Raw
hand
reinforce
Rgbs
Bbxs
handStat
bbf
bbf
gbbf
phi is infered after segmentation
simulation
inference
simulator
sample
task2
after segmentation

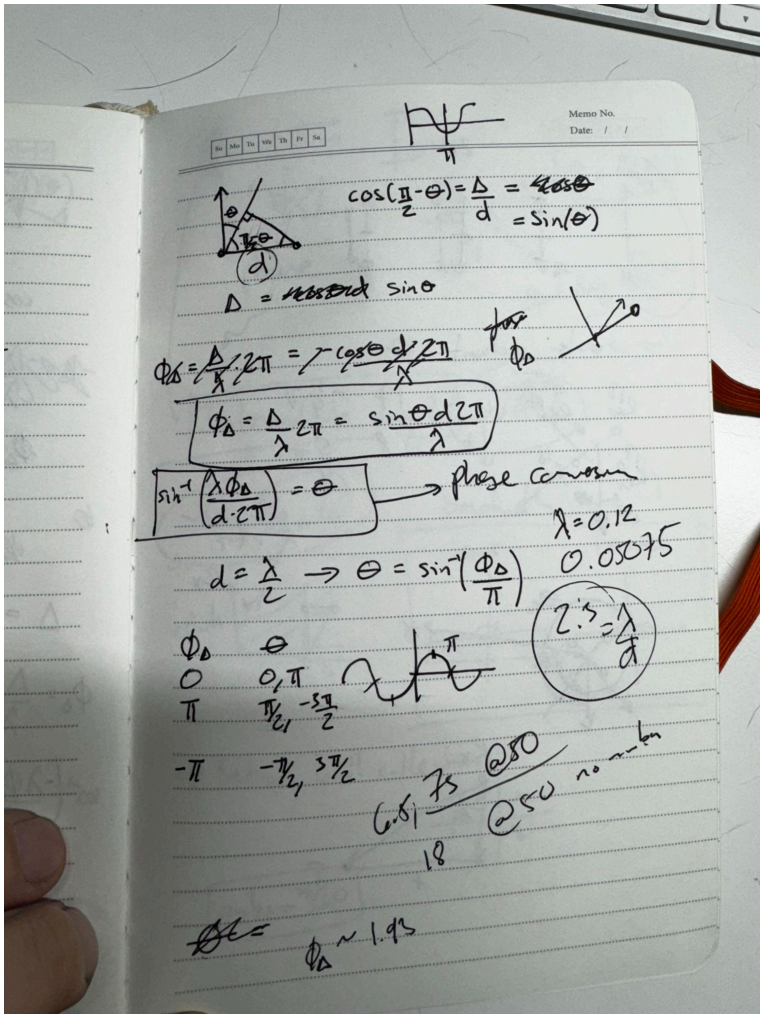
May 29

BATCH NORM HAS A WEIRD AFFECT ON CONVERGENCE, makes it asymmetric and weird

KIND OF WORKING
python simple_train.py -d
/home/ubuntu/data/wallarrayv3_2024_05_17_17_50_39_nRX2_rx_circle.zarr
/home/ubuntu/data/wallarrayv3_2024_05_18_00_55_58_nRX2_bounce.zarr
/home/ubuntu/data/wallarrayv3_2024_05_18_07_53_17_nRX2_bounce.zarr
/home/ubuntu/data/wallarrayv3_2024_05_18_16_29_28_nRX2_rx_circle.zarr
/home/ubuntu/data/wallarrayv3_2024_05_18_23_32_12_nRX2_bounce.zarr --device cuda
--batch 64 --workers 28 --type direct --lr 0.001 --shuffle --segmentation-level downsampled
--epochs 3 --depth 5 --hidden 128 --act leaky --other --seg-net conv --skip-segmentation
--wandb-project may29run_WTF --batch-norm --block
ON COMMIT 3e6c79cf8b1dc8d1addf5cb4eb59d62c70f1bc31

python simple_train.py -d
/home/ubuntu/data/wallarrayv3_2024_05_17_17_50_39_nRX2_rx_circle.zarr
/home/ubuntu/data/wallarrayv3_2024_05_18_00_55_58_nRX2_bounce.zarr
/home/ubuntu/data/wallarrayv3_2024_05_18_07_53_17_nRX2_bounce.zarr
/home/ubuntu/data/wallarrayv3_2024_05_18_16_29_28_nRX2_rx_circle.zarr
/home/ubuntu/data/wallarrayv3_2024_05_18_23_32_12_nRX2_bounce.zarr --device cuda
--batch 64 --workers 28 --type direct --lr 0.001 --shuffle --segmentation-level downsampled
--epochs 3 --depth 5 --hidden 64 --act selu --other --sigmoid --seg-net conv --skip-segmentation
--wandb-project may29run_WTF

May 15
To phase to theta



May 12

lrs="0.001 0.01 0.00001 0.0001"; types="direct discrete"; for type in $types; do for lr in $lrs; do python simple_train.py -d ~/data/*.zarr --device cuda --batch 512 --workers 16 --type $type --lr $lr --symmetry --segmentation-level downsampled --epochs 10; done ; done

May 11

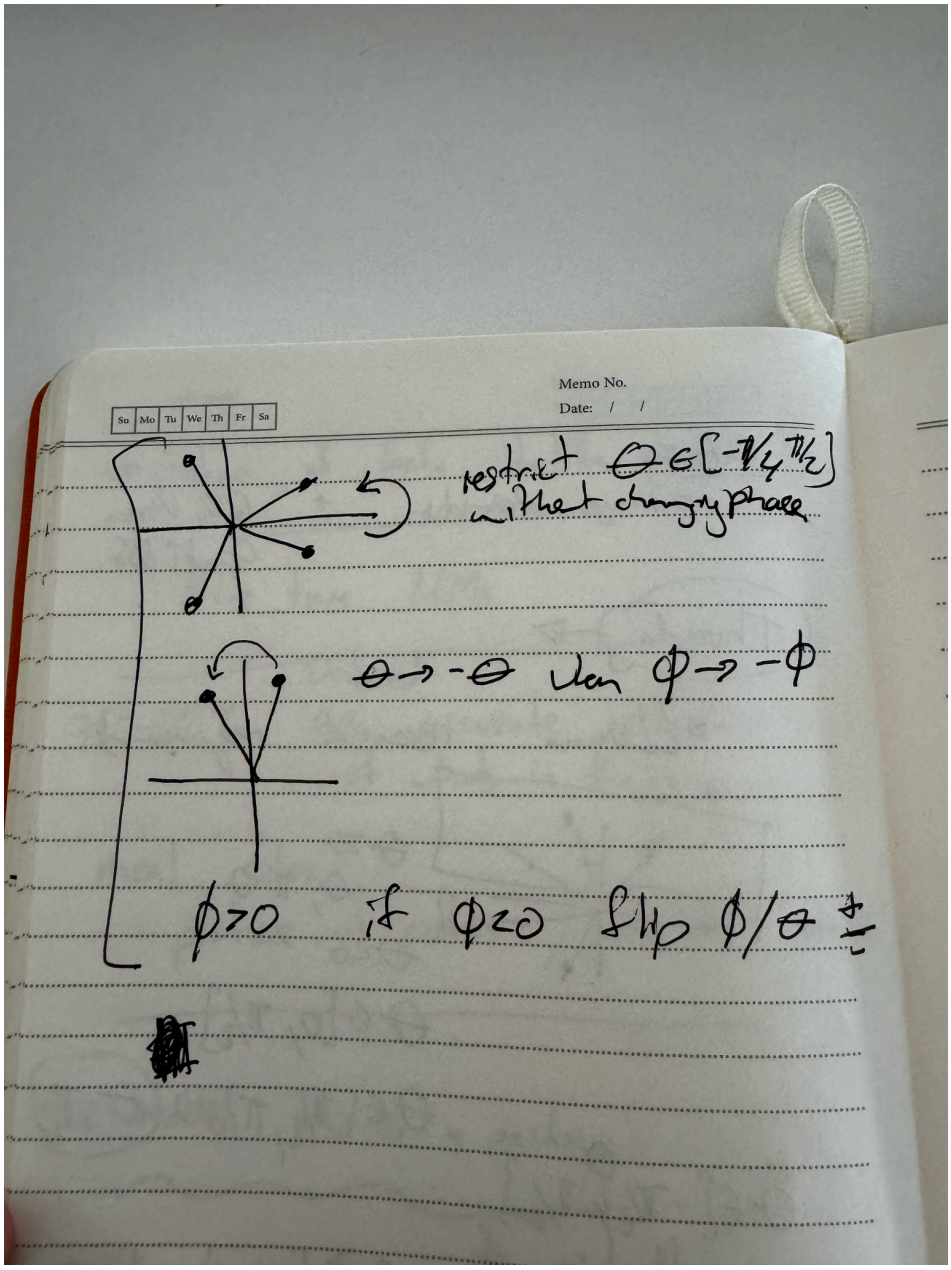
lrs="0.00001 0.0001 0.001"; types="direct discrete"; for type in $types; do for lr in $lrs; do python simple_train.py -d ~/wallarrayv3_2024_05_06_19_04_15_nRX2_bounce --device cuda --batch 48 --workers 16 --type $type --lr $lr; done ; done

Training segmentation network

python simple_train.py -d ~/wallarrayv3_2024_05_06_19_04_15_nRX2_bounce --device cuda --batch 32 --workers 16

python simple_train.py -d ~/wallarrayv3_2024_05_06_19_04_15_nRX2_bounce --device cuda --batch 32 --workers 16 --type direct

Symmetry



May 4

Test drone with docker python3 spf/mavlink_radio_collection.py -c spf/rover_configs/rover_emitter_config_pi.yaml -m /home/pi/device_mapping -r circle -t "RO2" -n 40000 --drone-uri tcp:192.168.1.133:14591

May 2

https://youtu.be/wUE8DI61s60

goulashparty.com

April 8:

If window is not guarded on both sides by 0 mean ish, then dont use that segment,

Brian todo:

Starter figure out how to get rover to not skid steer?
Prepare instructions for windows and code overview
Prepare instructions for simulator
Prepare instructions for simple network

March 24:

python spf/sdrpluto/sdr_controller.py --receiver-uri ip:192.168.1.18 --mode rx --rx-mode fast_attack --fc 5835000000 --rx-n 131072 --fs 16000000 --fi 3000000 --kernel-buffers 1

Captures DJI ^
https://www.youtube.com/watch?v=GxjVn7to3Ms
Aim for at least 2*131072 , 2*2**17 = 2**18

March 23:

https://github.com/plutoplus/plutoplus/issues/26
https://github.com/plutoplus/plutoplus/issues/26#issuecomment-1566127458

fw_setenv attr_name compatible
fw_setenv attr_val ad9361
fw_setenv compatible ad9361
fw_setenv mode 2r2t
reboot

AD9361 AD9364 AD9363
miskodzamba@Miskos-MBP ~ % iio_attr -C -u ip:192.168.1.17
IIO context with 9 attributes:
hw_model: Analog Devices PlutoSDR Rev.C (Z7010-AD9363A)
hw_model_variant: 1
hw_serial: 104000bac4950008230026001b440a003a
fw_version: v0.37-dirty
ad9361-phy,xo_correction: 40000000
ad9361-phy,model: ad9363a
local,kernel: 5.10.0-98725-g3eae70065be9-dirty
uri: ip:192.168.1.17
ip,ip-addr: 192.168.1.17

March 19

#docker build . -t ardupilotsim:v0.1
docker build . -t ghcr.io/misko/ardupilotsim:v0.1
docker push ghcr.io/misko/ardupilotsim:v0.1

March 18
https://stackoverflow.com/questions/75807997/start-docker-container-with-postgresql-as-a-fixture-in-conjunction-with-pytest-p

March 17
Mission with ok results!
https://discuss.ardupilot.org/t/ardupilot-and-docker-part-1/90532
https://hub.docker.com/u/ardupilot#
docker pull ardupilot/ardupilot-dev-base
docker pull ardupilot/ardupilot-dev-aarch64 (osx)

docker run --rm -it --entrypoint bash ardupilot/ardupilot-dev-armhf-musl

Build docker image

```
docker build . -t ardupilot
docker run -it -v `pwd`:/ardupilot ardupilot:latest bash
     git config --global --add safe.directory /ardupilot
     /ardupilot/Tools/gittools/submodule-sync.sh
```

https://aschmelyun.com/blog/using-docker-run-inside-of-github-actions/
/ardupilot/Tools/autotest/sim_vehicle.py -l 37.76509485,-122.40940127,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590
/ardupilot/Tools/autotest/sim_vehicle.py -L SpringValleyRoad -v rover -f rover-skid --out tcpin:0.0.0.0:14590

docker run --rm -it -p 14590:14590 ardupilot_spf bash

/ardupilot/modules/waf/waf-light build --target bin/ardurover
docker run --rm -it -p 14590:14590 ardupilot_spf /ardupilot/Tools/autotest/sim_vehicle.py -l 37.76509485,-122.40940127,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590

#build docker image , internally builds ardurover
#docker build --no-cache . -t ardupilot_spf
docker build . -t ardupilot_spf
docker run --rm -it -p 14590-14595:14590-14595 ardupilot_spf
/ardupilot/Tools/autotest/sim_vehicle.py -l 37.76509485,-122.40940127,0,0 -v rover -f rover-skid --out tcpin:0.0.0.0:14590 --out tcpin:0.0.0.0:14591 -S 5

param_fetch_all

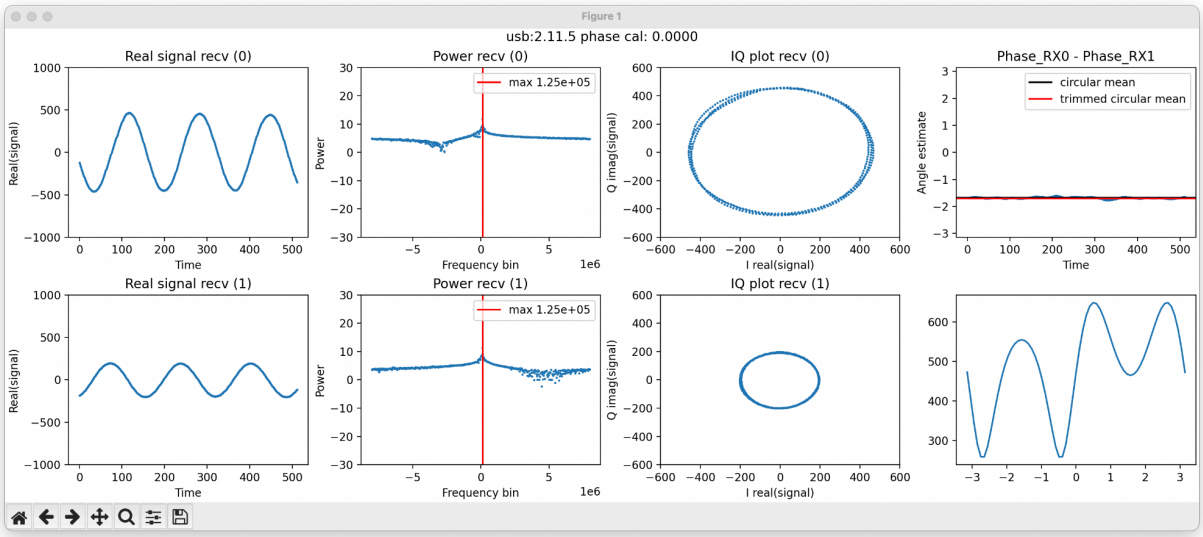
March 16
Smoother guided mode
Flash params on boot
Change compass to internal
Add tests
https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_Notify/ToneAlarm.cpp
Sync GPS time to system
https://firmware.ardupilot.org/Tools/ToneTester/#MFT240L8%20O4aO5dc%20O4aO5dc%20O4aO5dc%20L16dcdcdcdc
https://ardupilot.org/dev/docs/setting-up-sitl-on-linux.html
Building SITL
- Waf is inside adrupilot
- Run ./waf configure
- python -m pip install empy==3.3.4
- Dronecangen

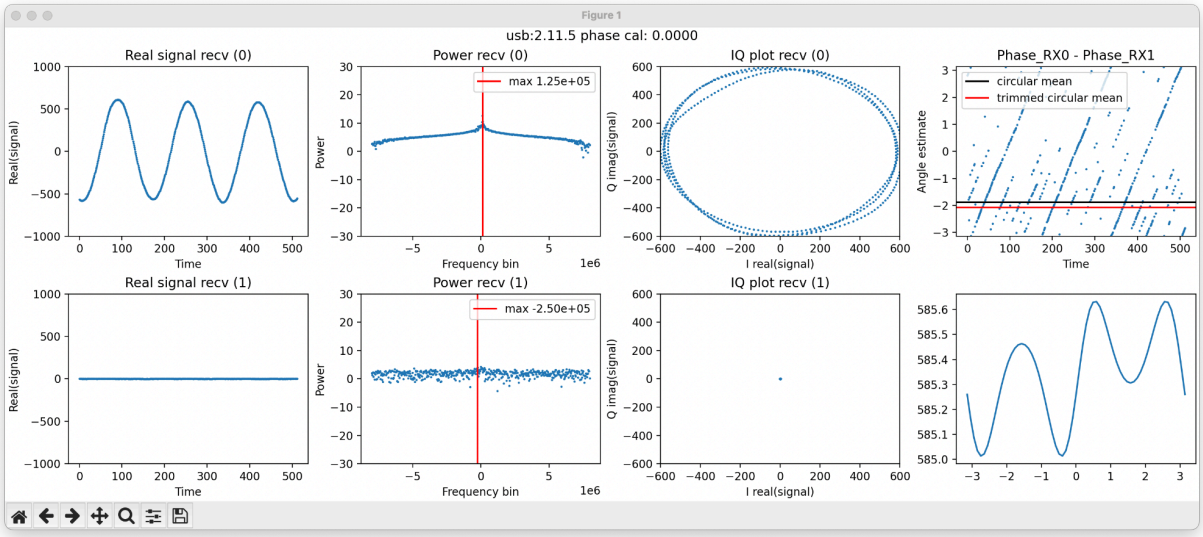
- Git clone https://github.com/ArduPilot/ardupilot.git
- Pip install pexpect dronecan
- python -m pip install empy==3.3.4
- Cd ardupilot [root repo]
- ./waf configure
- ./Tools/gittools/submodule-sync.sh

- python sim_vehicle.py -v rover -w
- python sim_vehicle.py -v rover -f rover-skid -l

March 10
https://github.com/mustafa-gokce/ardupilot-software-development/blob/main/mavproxy/automated-forwarding-services.md

March 9
- Field test
  - Ultrasonic not working
  - Bounces seem a bit off
  - RTL not working (seems to get hijacked by back to guided)
    - Manual also got hijacked





^ Fast attack vs slow attack (bottom). Fast attack seems much better!

March 8

Identify puts the board into a weird mode, need to reboot after

Raspberry pi 4 power cycle

sudo apt-get install uhubctl

sudo uhubctl -a 0

sudo uhubctl -a 1

Flashing firmware

https://ardupilot.org/dev/docs/bootloader.html

#https://firmware.ardupilot.org/Rover/stable-4.4.0/Pixhawk4/ardurover.apj

https://firmware.ardupilot.org/Rover/stable-4.4.0/fmuv3/ardurover.apj

wget https://raw.githubusercontent.com/ArduPilot/ardupilot/master/Tools/scripts/uploader.py

March 5

- https://mavlink.io/en/messages/ardupilotmega.html
- Dialect of adrdupilot mega
- Found gps time inside SYSTEM_TIME message
- Ultrasonic sensor online
- Designed and printing new front top with inserts for fans

March 3

- Fix image links x
- Fix aux links including pdf and gif from dropbox x
- Move data storage from spf dir to spf_aux x
- Move current spf dir to archive x
- Clone new spf x
- Avoidance sensor
  - How to mount on 3D
  - How to squeeze wires through
- Time sync
- Log compass , heading, and GPS

March 2

- Charge controller battery x
- Charge batteries
- Wall array
  - Pull data x
  - RX calibration x
  - Check RX calibration
- Pull settings from old rover x
- Prepare pi firmware
- Modify collector
  - Collect compass and GPS
  - Time sync
- Per rover
  - Flash pixhawks
  - Test motors

Finish battery harness
Mount avoidance sensor
Flash PI
Bond controller
Bond telemetry (pick different nets)
Run compass calibration
Finish skid plate mounts

Feb 25

Collect compass bearing inside of collection
Prepare franklin mission
Prepare new pi4
Flash
Test with usb collection

Feb 24

First guided point in simulator
GPS angle between points works

Todo:

Bounce in GPS
Can we assume lat and long are linear to distance?
Check wall array data
Start training new network
Print super base for use with USB connectors
High speed sampling over USB to be test

Goal for today:

Get GPS bounce working in simulator
Print next version of super base and rammer

Feb 23

Ordered USB connectors
Re designed to used fixed 10-32v -> 5v USB connectors
No need to solder custom connectors
Remove phase calibration in favor of simpler once TX on stays on!

Feb 13

Printed esp32 holders
Reversed esp32 protocol
Started wifi data collection

GPS 23° 27′ 30″ E.

23 deg, 27 minutes, 30 seconds

Minutes and seconds are base 60, sometimes seconds can be fractional

Sometimes just use minutes with decimals

Would be equal to 23° 27.5′ E

Sometimes just use degrees

23.45833° E

23+27/60+30/3600

Feb 11

Commit changes to github

Design esp32 holders

Get python mavlink online

https://mavlink.io/en/mavgen_python/

https://www.ardusub.com/developers/pymavlink.html

Message types:

https://mavlink.io/en/messages/common.html#GLOBAL_POSITION_INT

https://mavlink.io/kr/messages/common.html#MAV_FRAME

https://ardupilot.org/copter/docs/common-mavlink-mission-command-messages-mav_cmd.html

- Navigation commands are used to control the movement of the vehicle, including takeoff, moving to and around waypoints, changing altitude, and landing.
- DO commands are for auxiliary functions and do not affect the vehicle's position (for example, setting the camera trigger distance, or setting a servo value).
- Condition commands are used to delay DO commands until some condition is met, for example, the UAV reaches a certain altitude or distance from the waypoint.

https://ardupilot.org/copter/docs/common-mavlink-mission-command-messages-mav_cmd.html#mav-cmd-nav-waypoint

https://mavlink.io/en/messages/common.html#HOME_POSITION

https://mavlink.io/en/messages/common.html#MAV_CMD_REQUEST_MESSAGE
https://github.com/davidbuzz/ardupilot/blob/7292c89766b294e1280662392ab5f1cf6a442f8f/Tools/autotest/rover.py#L6578
https://discuss.bluerobotics.com/t/accessing-real-time-mavlink-messages-via-pymavlink/11355/14

ARM
https://www.youtube.com/watch?v=6M7e7DDLTQc

SAFETY?
https://discuss.ardupilot.org/t/switch-on-off-hardware-switch-using-pymavlink/65446

Hardware safety switch
https://discuss.ardupilot.org/t/safety-switch-by-mavlink/81152

Feb 10

ESP32 wifi testing

Windows machine

Installed silabs drivers

Connected and flashed RFTest190

Used rx listen

python sdr_controller.py --receiver-ip 192.168.1.16 --emitter-ip 192.168.1.16 --mode rx --tx-gain -25 --rx-spacing 0.05075 --fc 2412000000

When i hit start test in ESP32 it shows up!

Wrote ESP32 to firmware restarted, reopened and everything worked!

ESP tool writes “wifi tx continuous test!”

https://esp32.com/viewtopic.php?t=31116

To develop on XIAO board adding board manager

```
https://files.seeedstudio.com/arduino/package_seeeduino_boards_index.json
```

https://github.com/pothosware/SoapyPlutoSDR

Feb 5

https://github.com/Joshua-Riek/ubuntu-rockchip orange pi 5

Jan 26

Settable parameters:
https://wiki.analog.com/resources/tools-software/linux-drivers/iio-transceiver/ad9361

DC offset
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/542740/ad9361-reducing-rf-dc-offset-calibration-time

ENSM mode?
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/91510/ensm-mode-switching-mcs

FAQ
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/w/documents/18044/ad9361-faq

Gain tables

https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/w/documents/10071/gain-tables-in-csv-format

https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/w/documents/18045/ad936x-faq

Get gain setting

```
sdr._get_iio_attr('voltage0','hardwaregain', False)
```

Example RX/TX
https://pysdr.org/content/pluto.html

https://analogdevicesinc.github.io/pyadi-iio/buffers/index.html#buffer-examples
https://github.com/analogdevicesinc/pyadi-iio/blob/master/examples/pluto.py

https://github.com/analogdevicesinc/pyadi-iio/blob/master/examples/ad9361_example.py

https://ez.analog.com/adieducation/university-program/f/q-a/566149/pluto-sdr-pyadi-iio-continous-iq-recording

DC offset issues
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/78801/ad9361-rx-signal-inversation-when-gain-change
https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/100643/ad9361-rf-dc-offset-calibration-in-fast-lock-mode

Jan 25

RSSI

https://ez.analog.com/sw-interface-tools/f/q-a/549753/power-level-dbm-of-received-signal-from-adalm-pluto-receiver-using-pyadi-iio

rssi = sdr._ctrl.find_channel('voltage0').attrs['rssi'].value

```
sdr._ctrl.__dict__
for key in sdr._ctrl._debug_attrs:
   print(key)
```

MAP DEBUG ATTR to action

https://github.com/analogdevicesinc/linux/blob/88946d52e61d5b898c061d820caf27fd1c3730d7/drivers/iio/adc/ad9361.c#L8712

Some docs for phase inversion

https://github.com/analogdevicesinc/linux/blob/88946d52e61d5b898c061d820caf27fd1c3730d7/Documentation/devicetree/bindings/iio/adc/adi%2Cad9361.txt#L83

https://github.com/analogdevicesinc/linux/blob/main/drivers/iio/adc/ad9361.c

AD9361 SELF TEST INJECTION LOOPBACK

https://ez.analog.com/cfs-file/__key/communityserver-wikis-components-files/00-00-00-02-20/AD9361BISTFAQ.pdf

LNA BYPASS INVERT

https://github.com/analogdevicesinc/linux/blob/88946d52e61d5b898c061d820caf27fd1c3730d7/drivers/iio/adc/ad9361_regs.h#L780

https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/78801/ad9361-rx-signal-inversation-when-gain-change

```
/*
 *      REG_AGC_ATTACK_DELAY
 */
#define INVERT_BYPASSED_LNA_POLARITY          (1 << 6) /* Invert Bypassed LNA Polarity */
#define AGC_ATTACK_DELAY(x)                   (((x) & 0x3F) << 0) /* AGC Attack Delay<5:0> */
```

Where to set invert bits?

https://github.com/analogdevicesinc/linux/blob/main/drivers/iio/adc/ad9361.c#L3249

Jan24

Investigating random noise in wall array v2 estimates

https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/99352/ad9361-phase-difference-between-2-rx-channels

https://ez.analog.com/rf/wide-band-rf-transceivers/design-support/f/q-a/80250/ad9361-rx1-rx2-polarity-inversion

https://arrc.ou.edu/~cody/wni-radar/index.html

ALSO INVERTS AT LNA BYPASS!

https://analogdevicesinc.github.io/libiio/v0.25/python/index.html

How pyadiiio sets values

https://github.com/analogdevicesinc/pyadi-iio/blob/dbf2a606bf1dad4a803342c5641cc7ae61ff3282/adi/ad936x.py#L209

```
_control_device_name = "ad9361-phy"
_rx_data_device_name = "cf-ad9361-lpc"
_tx_data_device_name = "cf-ad9361-dds-core-lpc"
```

https://github.com/analogdevicesinc/libiio/blob/main/examples/ad9361-iiostream.c

#driver that details the rx1rx2_phase_inversion_en bit

https://github.com/analogdevicesinc/no-OS/blob/master/projects/ad9361/src/main.c

https://www.analog.com/media/en/technical-documentation/user-guides/ad9361.pdf

Links

Ardupilot

https://docs.google.com/document/d/1wb-RpmYjGr8dbCN2Y66siT5l7TH_BRIVrLd6BwdGB4g/edit

Libiio
https://docs.google.com/document/d/1W7VEFz4whHLrx4FWRhWrMVFuFcALknfa_ZKGog0kFp8/edit

GRBL setup
https://docs.google.com/document/d/1YIFWbLzY9jwrRHR3U6SyuFQ03p4VPgYUk4uI1-Ksbqk/edit

Signal processing
https://docs.google.com/document/d/1yeBeITCYm23qxHygSP66m4eWi-aXV9FPzmRflgPyxG4/edit

Project SPF high level
https://docs.google.com/document/d/1RUY4AjLBUgHPk1uudawPNFKCa6tr6kuIEzqYVg3OWG4/edit